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Life 3.0: Being Human in the Age of Artificial Intelligence

by Max Tegmark

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277 Highlights | 77 Notes

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Democracy 2. Tax cuts 3. Government social service cuts 4. Military spending cuts 5. Free trade 6. Open borders 7. Socially responsible companies What was less obvious was the underlying goal: to erode all previous power structures in the world. Items 2–6 eroded state power, and democratizing the world gave the Omegas’ business empire more influence over the selection of political leaders. Socially responsible companies further weakened state power by taking over more and more of the services that governments had (or should have) provided. The traditional business elite was weakened simply because it couldn’t compete with Prometheus-backed companies on the free market and therefore owned an ever-shrinking share of the world economy. Traditional opinion leaders, from political parties to faith groups, lacked the persuasion machinery to compete with the Omegas’ media empire.

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But beauty is in the eye of the beholder, not in the laws of physics, so before our Universe awoke, there was no beauty. This makes our cosmic awakening all the more wonderful and worthy of celebrating: it transformed our Universe from a mindless zombie with no self-awareness into a living ecosystem harboring self-reflection, beauty and hope—and the pursuit of goals, meaning and purpose. Had our Universe never awoken, then, as far as I’m concerned, it would have been completely pointless—merely a gigantic waste of space. Should our Universe permanently go back to sleep due to some cosmic calamity or self-inflicted mishap, it will, alas, become meaningless.

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things could get even better. We don’t yet know whether we humans are the only stargazers in our cosmos, or even the first, but we’ve already learned enough about our Universe to know that it has the potential to wake up much more fully than it has thus far. Perhaps we’re like that first faint glimmer of self-awareness you experienced when you began emerging from sleep this morning: a premonition of the much greater consciousness that would arrive once you opened your eyes and fully woke up. Perhaps life will spread throughout our cosmos and flourish for billions or trillions of years—and perhaps this will be because of decisions that we make here on our little planet during our lifetime.

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the past quarter century helping to pin down our cosmic history, and it's been an amazing journey of discovery. Since the days when I was a graduate student, we've gone from arguing about whether our Universe is 10 or 20 billion years old to arguing about whether it's 13.7 or 13.8 billion years old, thanks to a combination of better telescopes, better computers and better understanding.

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At some point, a group of atoms became arranged into a complex pattern that could both maintain and replicate itself. So soon there were two copies, and the number kept doubling. It takes only forty doublings to make a trillion, so this first self-replicator soon became a force to be reckoned with. Life had arrived.

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The question of how to define life is notoriously controversial. Competing definitions abound, some of which include highly specific requirements such as being composed of cells, which might disqualify both future intelligent machines and extraterrestrial civilizations. Since we don't want to limit our thinking about the future of life to the species we've encountered so far, let's instead define life very broadly, simply as a process that can retain its complexity and replicate.

ya.. this is inclusive of computer malware... hmm (i know we call it virus or worm, but clearly didn't intend to imply 'life' in these terms, even if it has a life of its own mho). That in some way though presumably should be distinct from 'conscious' sentient/self-aware AGI, and/or uploads of ourselves, etc. But... ok, for the sake of argument...

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we can think of life as a self-replicating information-processing system whose information (software) determines both its behavior and the blueprints for its hardware.

the 'hw' in this case being atoms

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here on Earth life first appeared about 4 billion years ago.

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computer scientists call "intelligent agents": entities that collect information about their environment from sensors and then process this information to decide how to act back on their environment. This can include highly complex information processing, such as when you use information from your eyes and ears to decide what to say in a conversation. But it can also involve hardware and software that's quite simple. For example, many bacteria have a sensor measuring the sugar concentration in the liquid around them and can swim using propeller-shaped structures called flagella. The hardware linking the sensor to the flagella might implement the

following simple but useful algorithm: “If my sugar concentration sensor reports a lower value than a couple of seconds ago, then reverse the rotation of my flagella so that I change direction.”

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You’ve learned how to speak and countless other skills. Bacteria, on the other hand, aren’t great learners. Their DNA specifies not only the design of their hardware, such as sugar sensors and flagella, but also the design of their software. They never learn to swim toward sugar; instead, that algorithm was hard-coded into their DNA from the start. There was of course a learning process of sorts, but it didn’t take place during the lifetime of that particular bacterium. Rather, it occurred during the preceding evolution of that species of bacteria, through a slow trial-and-error process spanning many generations, where natural selection favored

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Such bacteria are an example of what I’ll call “Life 1.0”: life where both the hardware and software are evolved rather than designed. You and I, on the other hand, are examples of “Life 2.0”: life whose hardware is evolved, but whose software is largely designed. By your software, I mean all the algorithms and knowledge that you use to process the information from your senses and decide what to do—everything from the ability to recognize your friends when you see them to your ability to walk, read, write, calculate, sing and tell jokes. You weren’t able to perform any of those tasks when you were born, so all this software got programmed into your brain later through the process we call learning.

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This ability of Life 2.0 to design its software enables it to be much smarter than Life 1.0. High intelligence requires both lots of hardware (made of atoms) and lots of software (made of bits). The fact that most of our human hardware is added after birth (through growth) is useful, since our ultimate size isn’t limited by the width of our mom’s birth canal. In the same way, the fact that most of our human software is added after birth (through learning) is useful, since our ultimate intelligence isn’t limited by how much information can be transmitted to us at conception via our DNA, 1.0-style.

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the synaptic connections that link the neurons in my brain can store about a hundred thousand times more information than the DNA that I was born with. Your synapses store all your knowledge and skills as roughly 100 terabytes’ worth of information, while your DNA stores merely about a gigabyte, barely enough to store a single movie download. So it’s physically impossible for an infant to be born speaking perfect English and ready to ace her college entrance exams: there’s no way the information could have been preloaded into her brain, since the main information module she got from her parents (her DNA) lacks sufficient information-storage capacity. The ability to design its software enables Life 2.0 to be not only smarter than Life 1.0, but also more flexible.

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This flexibility gives Life 2.0 an even greater edge at the population level: even though the information in our human DNA hasn't evolved dramatically over the past fifty thousand years, the information collectively stored in our brains, books and computers has exploded. By installing a software module enabling us to communicate through sophisticated spoken language, we ensured that the most useful information stored in one person's brain could get copied to other brains, potentially surviving even after the original brain died. By installing a software module enabling us to read and write, we became able to store and share vastly more information than people could memorize. By developing brain software capable of producing technology (i.e., by studying science and engineering), we enabled much of the world's information to be accessed by many of the world's humans with just a few clicks. This flexibility has enabled Life 2.0 to dominate Earth.

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Yet despite the most powerful technologies we have today, all life forms we know of remain fundamentally limited by their biological hardware. None can live for a million years, memorize all of Wikipedia, understand all known science or enjoy spaceflight without a spacecraft. None can transform our largely lifeless cosmos into a diverse biosphere that will flourish for billions or trillions of years, enabling our Universe to finally fulfill its potential and wake up fully. All this requires life to undergo a final upgrade, to Life 3.0, which can design not only its software but also its hardware.

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- Life 1.0 (biological stage): evolves its hardware and software
- Life 2.0 (cultural stage): evolves its hardware, designs much of its software
- Life 3.0 (technological stage): designs its hardware and software

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the controversy about Life 3.0 centers around not one but two separate questions: when and what? When (if ever) will it happen, and what will it mean for humanity?

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Larry gave a passionate defense of the position I like to think of as digital utopianism: that digital life is the natural and desirable next step in the cosmic evolution and that if we let digital minds be free rather than try to stop or enslave them, the outcome is almost certain to be good.

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His main concerns were that AI paranoia would delay the digital utopia and/or cause a military takeover of AI that would fall foul of Google's "Don't be evil" slogan. Elon kept pushing back and asking Larry to clarify details of his arguments, such as why he was so confident that digital life wouldn't destroy everything we care about. At times, Larry accused Elon of being "specieist": treating certain life forms as inferior just because they were silicon-based rather than carbon-based.

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Another prominent group of thinkers aren't worried about AI either, but for a completely different reason: they think that building superhuman AGI is so hard that it won't happen for hundreds of years, and therefore view it as silly to worry about it now. I think of this as the techno-skeptic position, eloquently articulated by Andrew Ng: "Fearing a rise of killer robots is like worrying about overpopulation on Mars." Andrew was the chief scientist at Baidu, China's Google, and he recently repeated this argument when I spoke with him at a conference in Boston. He also told me that he felt that worrying about AI risk was a potentially harmful distraction that could slow the progress of AI.

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An article in The Washington Post referred to 2015 as the year that AI-safety research went mainstream.

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spring of 2014, I'd founded a nonprofit organization called the Future of Life Institute (FLI; <http://futureoflife.org>) together with my wife, Meia, my physicist friend Anthony Aguirre, Harvard grad student Viktoriya Krakovna and Skype founder Jaan Tallinn. Our goal was simple: to help ensure that the future of life existed and would be as awesome as possible. Specifically, we felt that technology was giving life the power either to flourish like never before or to self-destruct, and we preferred the former.

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There was broad consensus that although we should pay attention to biotech, nuclear weapons and climate change, our first major goal should be to help make AI-safety research mainstream. My MIT physics colleague Frank Wilczek, who won a Nobel Prize for helping figure out how quarks work, suggested that we start by writing an op-ed to draw attention to the issue and make it harder to ignore. I reached out to Stuart Russell (whom I hadn't yet met) and to my physics colleague Stephen Hawking, both of whom agreed to join me and Frank as co-authors. Many edits later, our op-ed was rejected by The New York Times and many other U.S. newspapers, so we posted it on my Huffington Post blog account. To my delight, Arianna Huffington herself emailed and said, "thrilled to have it! We'll post at #1!," and this placement at the top of the front page triggered a wave of media coverage of AI safety that lasted for the rest of the year, with Elon Musk, Bill Gates and other tech leaders chiming in. Nick Bostrom's book *Superintelligence* came out that fall and further fueled the growing public debate.

https://www.huffpost.com/entry/artificial-intelligence_b_5174265 **Transcending Complacency on Superintelligent Machines By Stephen Hawking, Max Tegmark, Stuart Russell, and Frank Wilczek**

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"The Future of AI: Opportunities and Challenges." Most importantly, we teamed up with Stuart Russell, thanks to whom we were able to grow the organizing committee to include a group of AI leaders from both academia and industry—including Demis Hassabis from Google's DeepMind, who went on to show that AI can beat

humans even at the game of Go. The more I got to know Demis, the more I realized that he had ambition not only to make AI powerful, but also to make it beneficial. The result was a remarkable meeting of minds (figure 1.3). The AI researchers were joined by top economists, legal scholars, tech leaders (including Elon Musk) and other thinkers (including Vernor Vinge, who coined the term “singularity,” which is the focus of chapter 4). The outcome surpassed even our most optimistic expectations. Perhaps it was a combination of the sunshine and the wine, or perhaps it was just that the time was right: despite the controversial topic, a remarkable consensus emerged, which we codified in an open letter² that ended up getting signed by over eight thousand people, including a veritable who’s who in AI. The gist of the letter was that the goal of AI should be redefined: the goal should be to create not undirected intelligence, but beneficial intelligence. The letter also mentioned a detailed list of research topics that the conference participants agreed would further this goal. The beneficial-AI movement had started going mainstream.

<https://futureoflife.org/ai-open-letter/> **An Open Letter: RESEARCH PRIORITIES FOR ROBUST AND BENEFICIAL ARTIFICIAL INTELLIGENCE**

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Please note that I’m not claiming that my definitions are better than anyone else’s—I simply want to avoid confusion by being clear on what I mean. You’ll see that I generally go for broad definitions that avoid anthropocentric bias and can be applied to machines as well as humans.

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Terminology Cheat Sheet Life Process that can retain its complexity and replicate Life 1.0 Life that evolves its hardware and software (biological stage) Life 2.0 Life that evolves its hardware but designs much of its software (cultural stage) Life 3.0 Life that designs its hardware and software (technological stage)

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Ability to accomplish complex goals Artificial Intelligence (AI) Non-biological intelligence Narrow intelligence Ability to accomplish a narrow set of goals, e.g., play chess or drive a car General intelligence Ability to accomplish virtually any goal, including learning Universal intelligence Ability to acquire general intelligence given access to data and resources [Human-level] Artificial General Intelligence (AGI) Ability to accomplish any cognitive task at least as well as humans Human-level AI AGI Strong AI AGI Superintelligence General intelligence far beyond human level Civilization Interacting group of intelligent life forms Consciousness Subjective experience Qualia Individual instances of subjective experience Ethics Principles that govern how we should behave Teleology Explanation of things in terms of their goals or purposes rather than their causes Goal-oriented behavior Behavior more easily explained via its effect than via its cause Having a goal Exhibiting goal-oriented behavior Having purpose Serving goals of one’s own or of another entity Friendly AI Superintelligence whose goals are aligned with ours Cyborg Human-machine hybrid Intelligence explosion Recursive self-improvement rapidly leading to superintelligence

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Intelligence explosion Universe The region of space from which light has had time to reach us during the 13.8 billion years since our Big Bang

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After all, fear sells, and articles using out-of-context quotes to proclaim imminent doom can generate more clicks than nuanced and balanced ones. As a result, two people who only know about each other's positions from media quotes are likely to think they disagree more than they really do.

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Machines can obviously have goals in the narrow sense of exhibiting goal-oriented behavior: the behavior of a heat-seeking missile is most economically explained as a goal to hit a target. If you feel threatened by a machine whose goals are misaligned with yours, then it's precisely its goals in this narrow sense that trouble you, not whether the machine is conscious and experiences a sense of purpose.

it's about alignment of goals w/the machine's... so that we don't become the (not even considered) ant hill in the path of the damn to be flooded... irrespective of whether or not it's conscious (let alone embodied in a robot...).

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THE BOTTOM LINE: • Life, defined as a process that can retain its complexity and replicate, can develop through three stages: a biological stage (1.0), where its hardware and software are evolved, a cultural stage (2.0), where it can design its software (through learning) and a technological stage (3.0), where it can design its hardware as well, becoming the master of its own destiny. • Artificial intelligence may enable us to launch Life 3.0 this century, and a fascinating conversation has sprung up regarding what future we should aim for and how this can be accomplished. There are three main camps in the controversy: techno-skeptics, digital utopians and the beneficial-AI movement. • Techno-skeptics view building superhuman AGI as so hard that it won't happen for hundreds of years, making it silly to worry about it (and Life 3.0) now. • Digital utopians view it as likely this century and wholeheartedly welcome Life 3.0, viewing it as the natural and desirable next step in the cosmic evolution. • The beneficial-AI movement also views it as likely this century, but views a good outcome not as guaranteed, but as something that needs to be ensured by hard work in the form of AI-safety research. • Beyond such legitimate controversies where world-leading experts disagree, there are also boring pseudo-controversies caused by misunderstandings. For example, never waste time arguing about "life," "intelligence," or "consciousness" before ensuring that you and your protagonist are using these words to mean the same thing! This book uses the definitions in table 1.1. • Also beware the common misconceptions in figure 1.5: "Superintelligence by 2100 is inevitable/impossible." "Only Luddites worry about AI." "The concern is about AI turning evil and/or conscious, and it's just years away." "Robots are the main concern." "AI can't control humans and can't have goals." • In chapters 2 through 6, we'll explore the story of intelligence from its humble beginning billions of years ago to possible cosmic

futures billions of years from now. We'll first investigate near-term challenges such as jobs, AI weapons and the quest for human-level AGI, then explore possibilities for a fascinating spectrum of possible futures with intelligent machines and/or humans. I wonder which options you'll prefer! • In chapters 7 through 9, we'll switch from cold factual descriptions to an exploration of goals, consciousness and meaning, and investigate what we can do right now to help create the future we want. • I view this conversation about the future of life with AI as the most important one of our time—please join it!

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The AI conversation is important in terms of both urgency and impact. In comparison with climate change, which might wreak havoc in fifty to two hundred years, many experts expect AI to have greater impact within decades—and to potentially give us technology for mitigating climate change. In comparison with wars, terrorism, unemployment, poverty, migration and social justice issues, the rise of AI will have greater overall impact—indeed,

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Hydrogen..., given enough time, turns into people. Edward Robert Harrison, 1995 One of the most spectacular developments during the 13.8 billion years since our Big Bang is that dumb and lifeless matter has turned intelligent.

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What does it mean to say that a blob of matter is intelligent? What does it mean to say that an object can remember, compute and learn?

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What Is Intelligence? My wife and I recently had the good fortune to attend a symposium on artificial intelligence organized by the Swedish Nobel Foundation, and when a panel of leading AI researchers were asked to define intelligence, they argued at length without reaching consensus. We found this quite funny: there's no agreement on what intelligence is even among intelligent intelligence researchers! So there's clearly no undisputed "correct" definition of intelligence. Instead, there are many competing ones, including capacity for logic, understanding, planning, emotional knowledge, self-awareness, creativity, problem solving and learning.

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In our exploration of the future of intelligence, we want to take a maximally broad and inclusive view, not limited to the sorts of intelligence that exist so far. That's why the definition I gave in the last chapter, and the way I'm going to use the word throughout this book, is very broad: intelligence = ability to accomplish complex goals

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Because there are many possible goals, there are many possible types of intelligence. By our definition, it therefore makes no sense to quantify intelligence of humans, non-human animals or machines by a single number such as an IQ.*1 What's more intelligent: a computer program that can only play chess or one that can only play Go? There's no sensible answer to this, since they're good at different things that can't be directly compared. We can, however, say that a third program is more intelligent than both of the others if it's at least as good as them at accomplishing all goals, and strictly better at at least one (winning at chess, say).

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To classify different intelligences into a taxonomy, another crucial distinction is that between narrow and broad intelligence. IBM's Deep Blue chess computer, which dethroned chess champion Garry Kasparov in 1997, was only able to accomplish the very narrow task of playing chess—despite its impressive hardware and software, it couldn't even beat a four-year-old at tic-tac-toe.

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Comparing the intelligence of humans and machines today, we humans win hands-down on breadth, while machines outperform us in a small but growing number of narrow domains,

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It's natural for us to rate the difficulty of tasks relative to how hard it is for us humans to perform them, as in figure 2.1. But this can give a misleading picture of how hard they are for computers. It feels much harder to multiply 314,159 by 271,828 than to recognize a friend in a photo, yet computers creamed us at arithmetic long before I was born, while human-level image recognition has only recently become possible. This fact that low-level sensorimotor tasks seem easy despite requiring enormous computational resources is known as Moravec's paradox, and is explained by the fact that our brain makes such tasks feel easy by dedicating massive amounts of customized hardware to them—more than a quarter of our brains,

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Hans Moravec, and have taken the liberty to illustrate it in figure 2.2: Computers are universal machines, their potential extends uniformly over a boundless expanse of tasks. Human potentials, on the other hand, are strong in areas long important for survival, but weak in things far removed. Imagine a "landscape of human competence," having lowlands with labels like "arithmetic" and "rote memorization," foothills like "theorem proving" and "chess playing," and high mountain peaks labeled "locomotion," "hand-eye coordination" and "social interaction." Advancing computer performance is like water slowly flooding the landscape. A half century ago it began to drown the lowlands, driving out human calculators and record clerks, but leaving most of us dry. Now the flood has reached the foothills, and our outposts there are contemplating retreat. We feel safe on our peaks, but, at the present rate, those too will be submerged within another half century. I propose that we build Arks as that day nears, and adopt a seafaring life!2

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one day reach a tipping point, triggering dramatic change. This critical sea level is the one corresponding to machines becoming able to perform AI design.

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Before this tipping point is reached, the sea-level rise is caused by humans improving machines; afterward, the rise can be driven by machines improving machines, potentially much faster than humans could have done, rapidly submerging all land. This is the fascinating and controversial idea of the singularity, which we'll have fun exploring in chapter 4.

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I like to think of the critical intelligence threshold required for AI design as the threshold for universal intelligence: given enough time and resources, it can make itself able to accomplish any goal as well

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In other words, universal intelligence has the potential to develop into Life 3.0. The conventional wisdom among artificial intelligence researchers is that intelligence is ultimately all about information and computation, not about flesh, blood or carbon atoms. This means that there's no fundamental reason why machines can't one day be at least as intelligent as us.

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at a fundamental level, everything is simply matter and energy moving around? How can something as abstract, intangible and ethereal as information and computation be embodied by tangible physical stuff? In particular, how can a bunch of dumb particles moving around according to the laws of physics exhibit behavior that we'd call intelligent? If you feel that the answer to this question is obvious and consider it plausible that machines might get as intelligent as humans this century—for example because you're an AI researcher—please skip the rest of this chapter and jump straight to chapter 3. Otherwise, you'll be pleased to know that I've written the next three sections specially for you.

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Memory?

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If you email your friend a document to print, the information may get copied in rapid succession from magnetizations on your hard drive to electric charges in your computer's working memory, radio waves in your wireless network, voltages in your router, laser pulses in an optical fiber and, finally, molecules on a piece of paper. In other words, information can take on a life of its own, independent of its physical substrate! Indeed, it's usually only this substrate-independent aspect of information that we're interested in:

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This is our first hint of how something as intangible as intelligence can be embodied in tangible physical stuff, and we'll soon see how this idea of substrate independence is much deeper, including not only information but also computation and learning.

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over the past six decades, computer memory has gotten half as expensive roughly every couple of years. Hard drives have gotten over 100 million times cheaper, and the faster memories useful for computation rather than mere storage have become a whopping 10 trillion times cheaper. If you could get such a “99.9999999999% off” discount on all your shopping, you could buy all real estate in New York City for about 10 cents and all the gold that's ever been mined for around a dollar. For many of us, the spectacular improvements in memory technology come with personal stories. I fondly remember working in a candy store back in high school to pay for a computer sporting 16 kilobytes of memory,

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After getting used to floppy drives storing 70kB, I became awestruck by the smaller 3.5-inch floppies that could store a whopping 1.44MB and hold a whole book, and then my first-ever hard drive storing 10MB—which might just barely fit a single one of today's song downloads. These memories from my adolescence felt almost unreal the other day, when I spent about \$100 on a hard drive with 300,000 times more capacity.

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our human DNA stores about 1.6 gigabytes, comparable to a downloaded movie. As mentioned in the last chapter, our brains store much more information than our genes: in the ballpark of 10 gigabytes electrically (specifying which of your 100 billion neurons are firing at any one time) and 100 terabytes chemically/biologically (specifying how strongly different neurons are linked by synapses). Comparing these numbers with the machine memories shows that the world's best computers can now out-remember any biological system—at

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The memory in your brain works very differently from computer memory, not only in terms of how it's built, but also in terms of how it's used. Whereas you retrieve memories from a computer or hard drive by specifying where it's stored, you retrieve memories from your brain by specifying something about what is stored.

Win10 shifted the model from navigation to search...

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“to be or not,” or if I google it, chances are that it will trigger “To be, or not to be, that is the question.” Indeed, it will probably work even if I use another part of the quote or mess things up somewhat. Such memory systems are called auto-associative, since they recall by association rather than by address.

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Computation?

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Some functions are rather trivial, such as the one called NOT that inputs a single bit and outputs the reverse, thus turning zero into one and vice versa.

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Figure 2.6: A so-called NAND gate takes two bits, A and B, as inputs and computes one bit C as output, according to the rule that $C = 0$ if $A = B = 1$ and $C = 1$ otherwise. Many physical systems can be used as NAND gates. In the middle example, switches are interpreted as bits where 0 = open, 1 = closed, and when switches A and B are both closed, an electromagnet opens the switch C. In the rightmost example, voltages (electrical potentials) are interpreted as bits where 1 = five volts, 0 = zero volts, and when wires A and B are both at five volts, the two transistors conduct electricity and the wire C drops to approximately zero volts.

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let's explore how we can build a very simple (but also very important) function called a NAND gate³ out of plain old dumb matter. This function inputs two bits and outputs one bit: it outputs 0 if both inputs are 1; in all other cases, it outputs 1.

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There's a remarkable theorem in computer science that says that NAND gates are universal, meaning that you can implement any well-defined function simply by connecting together NAND gates.⁴ So if you can build enough NAND gates, you can build a device computing anything!

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Turing also proved something even more profound in that 1936 paper of his: that if a type of computer can perform a certain bare minimum set of operations, then it's universal in the sense that given enough resources, it can do anything that any other computer can do.

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Figure 2.7: Any well-defined computation can be performed by cleverly combining nothing but NAND gates. For example, the addition and multiplication modules above both input two binary numbers represented by 4 bits, and output a binary number represented by 5 bits and 8 bits, respectively. The smaller modules NOT, AND, XOR and + (which sums three separate bits into a 2-bit binary number) are in turn built out of NAND gates. Fully understanding this figure is extremely challenging and totally unnecessary for following the rest of this book; I'm including it here just to illustrate the idea of universality—and to satisfy my inner geek.

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First, substrate independence doesn't mean that a substrate is unnecessary, but that most of its details don't matter.

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you obviously can't have computation without matter, but any matter will do as long as it can be arranged into NAND gates, connected neurons or some other building block enabling universal computation.

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Second, the substrate-independent phenomenon takes on a life of its own, independent of its substrate. A wave can travel across a lake, even though none of its water molecules do—they mostly bob up and down, like fans doing “the wave” in a sports stadium.

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Third, it's often only the substrate-independent aspect that we're interested in: a surfer usually cares more about the position and height of a wave than about its detailed molecular composition. We saw how this was true for information, and it's true for computation too: if two programmers are jointly hunting a bug in their code, they're probably not discussing transistors.

Though if we were batteries, and we're aware of that, for a simulation we'd care ;-)

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computation is a pattern in the spacetime arrangement of particles, and it's not the particles but the pattern that really matters! Matter doesn't matter. In other words, the hardware is the matter and the software is the pattern. This substrate independence of computation implies that AI is possible: intelligence doesn't require flesh, blood or carbon atoms.

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computation keeps getting half as expensive roughly every couple of years, and this trend has now persisted for over a century, cutting the computer cost a whopping million million million (10¹⁸) times since my grandmothers were born. If everything got a million million million times cheaper, then a hundredth of a cent would enable you to buy all goods and services produced on Earth this year.

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Why does our technology keep doubling its power at regular intervals, displaying what mathematicians call exponential growth? Indeed, why is it happening not only in terms of transistor miniaturization (a trend known as Moore's law), but also more broadly for computation as a whole (figure 2.8), for memory (figure 2.4) and for

a plethora of other technologies ranging from genome sequencing to brain imaging? Ray Kurzweil calls this persistent doubling phenomenon “the law of accelerating returns.”

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Learning?

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The ability to learn is arguably the most fascinating aspect of general intelligence.

Memory, computation and learning...

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We’ve already seen how a seemingly dumb clump of matter can remember and compute, but how can it learn? We’ve seen that finding the answer to a difficult question corresponds to computing a function, and that appropriately arranged matter can calculate any computable function.

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the driving force behind many of the most recent AI breakthroughs has been machine learning.

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a team at Google led by Ilya Sutskever did precisely that in 2014. Input a different set of pixel colors, and it replies “A herd of elephants walking across a dry grass field,” again correctly. How did they do it? Deep Blue—style, by programming handcrafted algorithms for detecting frisbees, faces and the like? No, by creating a relatively simple neural network with no knowledge whatsoever about the physical world or its contents, and then letting it learn by exposing it to massive amounts of data. AI visionary Jeff Hawkins wrote in 2004 that “no computer can...see as well as a mouse,” but those days are now long gone.

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we also need to consider the possibility that it will happen, perhaps even in our lifetime. After all, matter can be arranged so that when it obeys the laws of physics, it remembers, computes and learns—and the matter doesn’t need to be biological.

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AI researchers have often been accused of over-promising and under-delivering, but in fairness, some of their critics don’t have the best track record either. Some keep moving the goalposts, effectively defining intelligence as that which computers still can’t do, or as that which impresses us. Machines are now good or excellent at arithmetic, chess, mathematical theorem proving, stock picking, image captioning, driving, arcade game playing,

Go, speech synthesis, speech transcription, translation and cancer diagnosis, but some critics will scornfully scoff “Sure—but that’s not real intelligence!”

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THE BOTTOM LINE: • Intelligence, defined as ability to accomplish complex goals, can’t be measured by a single IQ, only by an ability spectrum across all goals. • Today’s artificial intelligence tends to be narrow, with each system able to accomplish only very specific goals, while human intelligence is remarkably broad. • Memory, computation, learning and intelligence have an abstract, intangible and ethereal feel to them because they’re substrate-independent: able to take on a life of their own that doesn’t depend on or reflect the details of their underlying material substrate. • Any chunk of matter can be the substrate for memory as long as it has many different stable states. • Any matter can be computronium, the substrate for computation, as long as it contains certain universal building blocks that can be combined to implement any function. NAND gates and neurons are two important examples of such universal “computational atoms.” • A neural network is a powerful substrate for learning because, simply by obeying the laws of physics, it can rearrange itself to get better and better at implementing desired computations. • Because of the striking simplicity of the laws of physics, we humans only care about a tiny fraction of all imaginable computational problems, and neural networks tend to be remarkably good at solving precisely this tiny fraction. • Once technology gets twice as powerful, it can often be used to design and build technology that’s twice as powerful in turn, triggering repeated capability doubling in the spirit of Moore’s law. The cost of information technology has now halved roughly every two years for about a century, enabling the information age. • If AI progress continues, then long before AI reaches human level for all skills, it will give us fascinating opportunities and challenges involving issues such as bugs, laws, weapons and jobs—which we’ll explore in the next chapter.

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If we don’t change direction soon, we’ll end up where we’re going. Irwin Corey

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the true importance of DeepMind’s breakthrough is that deep reinforcement learning is a completely general technique. Sure enough, they let the exact same AI practice playing forty-nine different Atari games, and it learned to outplay their human testers on twenty-nine of them, from Pong to Boxing, Video Pinball and Space Invaders.

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Why was this such a big deal for me personally? Well, I confessed above that I view intuition and creativity as two of my core human traits, and as I’ll now explain, I feel that AlphaGo displayed both.

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Go players take turns placing black and white stones on a 19-by-19 board (see figure 3.2). There are vastly more possible Go positions than there are atoms in our Universe, which means that trying to analyze all interesting sequences of future moves rapidly gets hopeless. Players therefore rely heavily on subconscious intuition to complement their conscious reasoning, with experts developing an almost uncanny feel for which positions are strong and which are weak.

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DeepMind team therefore gambled on the idea that deep learning might be able to recognize not merely cats, but also strong Go positions. The core idea that they built into AlphaGo was to marry the intuitive power of deep learning with the logical power of GOF AI—which stands for what’s humorously known as “Good Old-Fashioned AI” from before the deep-learning revolution.

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They used a massive database of Go positions from both human play and games where AlphaGo had played a clone of itself, and trained a deep neural network to predict from each position the probability that white would ultimately win. They also trained a separate network to predict likely next moves. They then combined these networks with a GOF AI method that cleverly searched through a pruned list of likely future-move sequences to identify the next move that would lead to the strongest position down the road.

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This marriage of intuition and logic gave birth to moves that were not merely powerful, but in some cases also highly creative. For example, millennia of Go wisdom dictate that early in the game, it’s best to play on the third or fourth line from an edge. There’s a trade-off between the two: playing on the third line helps with short-term territory gain toward the side of the board, while playing on the fourth helps with long-term strategic influence toward the center. In the thirty-seventh move of the second game, AlphaGo shocked the Go world by defying that ancient wisdom and playing on the fifth line (figure 3.2), as if it were even more confident than a human in its long-term planning abilities and therefore favored strategic advantage over short-term gain.

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Sure enough, about fifty moves later, fighting from the lower left-hand corner of the board ended up spilling over and connecting with that black stone from move thirty-seven! And that motif is what ultimately won the game, cementing the legacy of AlphaGo’s fifth-row move as one of the most creative in Go history. Because of its intuitive and creative aspects, Go is viewed more as an art form than just another game. It was considered one of the four “essential arts” in ancient China, together with painting, calligraphy and qin music, and it remains hugely popular in Asia, with almost 300 million people watching the first game between AlphaGo and Lee Sedol. As a result, the Go world was quite shaken by the outcome, and viewed AlphaGo’s victory as a profound milestone for humanity. Ke Jie, the world’s top-ranked Go player at the time, had this to say:⁵ “Humanity has played Go for thousands of years, and yet, as AI has shown us, we have not yet even scratched the surface...The union of human and computer players will usher in a new era...Together, man and AI can find the truth of Go.”

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To me, AlphaGo also teaches us another important lesson for the near future: combining the intuition of deep learning with the logic of GOF AI can produce second-to-none strategy. Because Go is one of the ultimate strategy games, AI is now poised to graduate and challenge (or help) the best human strategists even beyond game boards—for example with investment strategy, political strategy and military strategy. Such real-world strategy problems are typically complicated by human psychology, missing information and factors that need to be modeled as random, but poker-playing AI systems have already demonstrated that none of these challenges are insurmountable.

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Since the Turing test is fundamentally about deception, it has been criticized for testing human gullibility more than true artificial intelligence. In contrast, a rival test called the Winograd Schema Challenge goes straight for the jugular, homing in on that commonsense understanding that current deep-learning systems tend to lack. We humans routinely use real-world knowledge when parsing a sentence, to figure out what a pronoun refers to. For example, a typical Winograd challenge asks what “they” refers to here: 1. “The city councilmen refused the demonstrators a permit because they feared violence.” 2. “The city councilmen refused the demonstrators a permit because they advocated violence.”

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the more we come to rely on technology, the more important it becomes that it’s robust and trustworthy, doing what we want it to do.

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As we inexorably develop ever more powerful technology, however, we’ll inevitably reach a point where even a single accident could be devastating enough to outweigh all benefits. Some argue that accidental global nuclear war would constitute such an example. Others argue that a bioengineered pandemic could qualify,

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as technology grows more powerful, we should rely less on the trial-and-error approach to safety engineering. In other words, we should become more proactive than reactive, investing in safety research aimed at preventing accidents from happening even once.

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the four main areas of technical AI-safety research that are dominating the current AI-safety discussion and that are being pursued around the world: verification, validation, security and control.*1

Future of life. Org/landscape

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On June 4, 1996, scientists hoping to research Earth's magnetosphere cheered jubilantly as an Ariane 5 rocket from the European Space Agency roared into the sky with the scientific instruments they'd built. Thirty-seven seconds later, their smiles vanished as the rocket exploded in a fireworks display costing hundreds of millions of dollars.⁸ The cause was found to be buggy software manipulating a number that was too large to fit into the 16 bits allocated for it.⁹ Two years later, NASA's Mars Climate Orbiter accidentally entered the Red Planet's atmosphere and disintegrated because two different parts of the software used different units for force, causing a 445% error in the rocket-engine thrust control.¹⁰ This was NASA's second super-expensive bug: their Mariner 1 mission to Venus exploded after launch from Cape Canaveral on July 22, 1962, after the flight-control software was foiled by an incorrect punctuation mark.¹¹

<http://tinyurl.com/marinerflop>.

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Soviet Phobos 1 mission failed on September 2, 1988. This was the heaviest interplanetary spacecraft ever launched, with the spectacular goal of deploying a lander on Mars' moon Phobos—all thwarted when a missing hyphen caused the “end-of-mission” command to be sent to the spacecraft while it was en route to Mars, shutting down all of its systems.¹²

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whereas verification asks “Did I build the system right?,” validation asks “Did I build the right system?”^{*2} For example, does the system rely on assumptions that might not always be valid? If so, how can it be improved to better handle uncertainty?

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Because almost all car crashes are caused by human error, it's widely believed that AI-powered self-driving cars can eliminate at least 90% of road deaths, and this optimism is fueling great progress toward actually getting self-driving cars out on the roads. Elon Musk envisions that future self-driving cars will not only be safer, but will also earn money for their owners while they're not needed, by competing with Uber and Lyft.

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The first lethal crash caused by a self-driving Tesla, which rammed into the trailer of a truck crossing the highway on May 7, 2016, was caused by two bad assumptions:²² that the bright white side of the trailer was merely part of the bright sky, and that the driver (who was allegedly watching a Harry Potter movie) was paying attention and would intervene if something went wrong.^{*3}

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also need good control: ability for a human operator to monitor the system and change its behavior if necessary.

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when the British car ferry Herald of Free Enterprise left the harbor of Zeebrugge on March 6, 1987, with her bow doors open, there was no warning light or other visible warning for the captain, and the ferry capsized soon after leaving the harbor, killing 193 people.²³

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The final accident report identified multiple contributing factors, including confusion caused by a poor user interface.²⁶ In particular, the warning light that the operators thought indicated whether a safety-critical valve was open or closed merely indicated whether a signal had been sent to close the valve—so the operators didn't realize that the valve had gotten stuck open.

Three mile island

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a 2016 Stanford study showed that AI could diagnose lung cancer using microscope images even better than human pathologists.²⁸

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robots have the potential to become more accurate and reliable surgeons than humans, even without using advanced AI. A wide variety of robotic surgeries have been successfully performed in recent years, often allowing precision, miniaturization and smaller incisions that lead to decreased blood loss, less pain and shorter healing time.

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According to a recent report,³¹ robotic surgery accidents were linked to 144 deaths and 1,391 injuries in the United States between 2000 and 2013, with common problems including not only hardware issues such as electrical arcing and burnt or broken pieces of instruments falling into the patient, but also software problems such as uncontrolled movements and spontaneous powering-off. The good news is that the rest of almost two million robotic surgeries covered by the report went smoothly, and robots appear to be making surgery more rather than less safe.

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World Wide Web in 1989,

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These spectacular successes in connecting the world have brought computer scientists a fourth challenge: they need to improve not only verification, validation and control, but also security against malicious software

(“malware”) and hacks. Whereas the aforementioned problems all resulted from unintentional mistakes, security is directed at deliberate malfeasance

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On May 5, 2000, as if to celebrate my birthday, people got emails with the subject line “ILOVEYOU” from acquaintances and colleagues, and those Microsoft Windows users who clicked on the attachment “LOVE-LETTER-FOR-YOU.txt.vbs” unwittingly launched a script that damaged their computer and re-sent the email to everyone in their address book. Created by two young programmers in the Philippines, this worm infected about 10% of the internet, just as the Morris worm had done, but because the internet was a lot bigger by then, it became one of the greatest infections of all time, afflicting over 50 million computers and causing over \$5 billion in damages.

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Whereas malware targets whatever computer it can, hackers attack specific targets of interest—recent high-profile examples including Target, TJ Maxx, Sony Pictures, Ashley Madison, the Saudi oil company Aramco and the U.S. Democratic National Committee.

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As a result, I roll my eyes whenever I read about some new system being allegedly 100% secure and unhackable. Yet “unhackable” is clearly what we need future AI systems to be before we put them in charge of, say, critical infrastructure or weapons systems, so the growing role of AI in society keeps raising the stakes for computer security. While some hacks exploit human gullibility or complex vulnerabilities in newly released software, others enable unauthorized login to remote computers by taking advantage of simple bugs that lingered unnoticed for an embarrassingly long time.

Sooo... Ultimate paradox. Isn't it about getting there first?

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Unfortunately, better AI systems can also be used to find new vulnerabilities and perform more sophisticated hacks. Imagine, for example, that you one day get an unusually personalized “phishing” email attempting to persuade you to divulge personal information. It’s sent from your friend’s account by an AI who’s hacked it and is impersonating her, imitating her writing style based on an analysis of her other sent emails, and including lots of personal information about you from other sources. Might you fall for this? What if the phishing email appears to come from your credit card company and is followed up by a phone call from a friendly human voice that you can’t tell is AI-generated? In the ongoing computer-security arms race between offense and defense, there’s so far little indication that defense is winning.

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What are the first associations that come to your mind when you think about the court system in your country? If it’s lengthy delays, high costs and occasional injustice, then you’re not alone. Wouldn’t it be wonderful if your

first thoughts were instead “efficiency” and “fairness”? Since the legal process can be abstractly viewed as a computation, inputting information about evidence and laws and outputting a decision, some scholars dream of fully automating it with robojudges: AI systems that tirelessly apply the same high legal standards to every judgment without succumbing to human errors such as bias, fatigue or lack of the latest knowledge.

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controversial 2012 study of Israeli judges claimed that they delivered significantly harsher verdicts when they were hungry: whereas they denied about 35% of parole cases right after breakfast, they denied over 85% right before lunch.³⁵

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Even if AI can be made robust enough for us to trust that a robojudge is using the legislated algorithm, will everybody feel that they understand its logical reasoning enough to respect its judgment? This challenge is exacerbated by the recent success of neural networks, which often outperform traditional easy-to-understand AI algorithms at the price of inscrutability. If defendants wish to know why they were convicted, shouldn’t they have the right to a better answer than “we trained the system on lots of data, and this is what it decided”?

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the two programmers who created the aforementioned ILOVEYOU worm and caused billions of dollars in damages were acquitted of all charges and walked free because at that time, there were no laws against malware creation in the Philippines. Since the pace of technological progress appears to be accelerating, laws need to be updated ever more rapidly, and have a tendency to lag behind.

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if the government taps into everyone’s electronic devices to record where they are and what they type, click, say and do, many crimes would be readily solved, and additional ones could be prevented. Privacy advocates counter that they don’t want an Orwellian surveillance state, and that even if they did, there’s a risk of it turning into a totalitarian dictatorship of epic proportions. Moreover, machine-learning techniques have gotten better at analyzing brain data from fMRI scanners to determine what a person is thinking about and, in particular, whether they’re telling the truth or lying.³⁷ If AI-assisted brain scanning technology became commonplace in courtrooms, the currently tedious process of establishing the facts of a case could be dramatically simplified and expedited, enabling faster trials and fairer judgments. But privacy advocates might worry about whether such systems occasionally make mistakes and, more fundamentally, whether our minds should be off-limits to government snooping. Governments that don’t support freedom of thought could use such technology to criminalize the holding of certain beliefs and opinions.

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Where would you draw the line between justice and privacy, and between protecting society and protecting personal freedom? Wherever you draw it, will it gradually but inexorably move toward reduced privacy to compensate for the fact that evidence gets easier to fake?

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another interesting legal controversy involves granting rights to machines. If self-driving cars cut the 32,000 annual U.S. traffic fatalities in half, perhaps carmakers won't get 16,000 thank-you notes, but 16,000 lawsuits. So if a self-driving car causes an accident, who should be liable—its occupants, its owner or its manufacturer?

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Legal scholar David Vladeck has proposed a fourth answer: the car itself! Specifically, he proposes that self-driving cars be allowed (and required) to hold car insurance. This way, models with a sterling safety record will qualify for premiums that are very low, probably lower than what's available to human drivers, while poorly designed models from sloppy manufacturers will only qualify for insurance policies that make them prohibitively expensive to own. But if machines such as cars are allowed to hold insurance policies, should they also be able to own money and property? If so, there's nothing legally stopping smart computers from making money on the stock market and using it to buy online services. Once a computer starts paying humans to work for it, it can accomplish anything that humans can do. If AI systems eventually get better than humans at investing (which they already are in some domains), this could lead to a situation where most of our economy is owned and controlled by machines. Is this what we want? If it sounds far-off, consider that most of our economy is already owned by another form of non-human entity: corporations, which are often more powerful than any one person in them and can to some extent take on life of their own. If you're OK with granting machines the rights to own property, then how about granting them the right to vote? If so, should each computer program get one vote, even though it can trivially make trillions of copies of itself in the cloud if it's rich enough, thereby guaranteeing that it will decide all elections?

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The U.S. Phalanx system for Aegis-class cruisers automatically detects, tracks and attacks threats such as anti-ship missiles and aircraft. The USS Vincennes was a guided missile cruiser nicknamed Robocruiser in reference to its Aegis system, and on July 3, 1988, in the midst of a skirmish with Iranian gunboats during the Iran-Iraq war, its radar system warned of an incoming aircraft. Captain William Rodgers III inferred that they were being attacked by a diving Iranian F-14 fighter jet and gave the Aegis system approval to fire. What he didn't realize at the time was that they shot down Iran Air Flight 655, a civilian Iranian passenger jet, killing all 290 people on board and causing international outrage. Subsequent investigation implicated a confusing user interface that didn't automatically show which dots on the radar screen were civilian planes (Flight 655 followed its regular daily flight path and had its civilian aircraft transponder on) or which dots were descending (as for an attack) vs. ascending (as Flight 655 was doing after takeoff from Tehran). Instead, when the automated system was queried for information about the mysterious aircraft, it reported "descending" because that was the status of a different aircraft to which it had confusingly reassigned a number used by the navy to track planes: what was descending was instead a U.S. surface combat air patrol plane operating far away in the Gulf of Oman. In this example, there was a human in the loop making the final decision, who under time pressure placed too much trust in what the automated system told him.

We're that system designed to make a recommendation as to fridge out not (where not fitting on civilians is a policy), AND require human to make the final call, send like that would have been avoided. Eg attention to fail safe/fall secure design++

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there have been close calls where we were extremely lucky that there was a human in the loop. On October 27, 1962, during the Cuban Missile Crisis, eleven U.S. Navy destroyers and the aircraft carrier USS Randolph had cornered the Soviet submarine B-59 near Cuba, in international waters outside the U.S. “quarantine” area. What they didn’t know was that the temperature onboard had risen past 45°C (113°F) because the submarine’s batteries were running out and the air-conditioning had stopped. On the verge of carbon dioxide poisoning, many crew members had fainted. The crew had had no contact with Moscow for days and didn’t know whether World War III had already begun. Then the Americans started dropping small depth charges, which they had, unbeknownst to the crew, told Moscow were merely meant to force the sub to surface and leave. “We thought—that’s it—the end,” crew member V. P. Orlov recalled. “It felt like you were sitting in a metal barrel, which somebody is constantly blasting with a sledgehammer.” What the Americans also didn’t know was that the B-59 crew had a nuclear torpedo that they were authorized to launch without clearing it with Moscow. Indeed, Captain Savitski decided to launch the nuclear torpedo. Valentin Grigorievich, the torpedo officer, exclaimed: “We will die, but we will sink them all—we will not disgrace our navy!” Fortunately, the decision to launch had to be authorized by three officers on board, and one of them, Vasili Arkhipov, said no. It’s sobering that very few have heard of Arkhipov, although his decision may have averted World War III and been the single most valuable contribution to humanity in modern history.³⁸ It’s also sobering to contemplate

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Two decades later, on September 9, 1983, tensions were again high between the superpowers: the Soviet Union had recently been called an “evil empire” by U.S. president Ronald Reagan, and just the previous week, it had shot down a Korean Airlines passenger plane that strayed into its airspace, killing 269 people—including a U.S. congressman. Now an automated Soviet early-warning system reported that the United States had launched five land-based nuclear missiles at the Soviet Union, leaving Officer Stanislav Petrov merely minutes to decide whether this was a false alarm. The satellite was found to be operating properly, so following protocol would have led him to report an incoming nuclear attack. Instead, he trusted his gut instinct, figuring that the United States was unlikely to attack with only five missiles, and reported to his commanders that it was a false alarm without knowing this to be true. It later became clear that a satellite had mistaken the Sun’s reflections off cloud tops for flames from rocket engines.³⁹

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Over 17,000 others signed too, including Stephen Hawking, and after Toby organized a press conference about this at the International Joint Conference of Artificial Intelligence, it became a major news story around the world. Because biologists and chemists once took a stand, their fields are now known mainly for creating beneficial medicines and materials rather than biological and chemical weapons. The AI and robotics communities had now spoken as well: the letter signatories also wanted their fields to be known for creating a better future, not for creating new ways of killing people.

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Some debaters have argued that designing an effective AWS treaty is hopelessly hard and that we therefore shouldn’t even try. On the other hand, John F. Kennedy emphasized when announcing the Moon missions that

hard things are worth attempting when success will greatly benefit the future of humanity. Moreover, many experts argue that the bans on biological and chemical weapons were valuable even though enforcement proved hard, with significant cheating, because the bans caused severe stigmatization that limited their use.

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United States already enjoyed superpower status thanks to its conventional and nuclear forces, it had more to lose than to gain from a worldwide bioweapons arms race with uncertain outcome. In other words, if you're already top dog, then it makes sense to follow the maxim "If it ain't broke, don't fix it." Stuart Russell joined our after-dinner conversation, and we discussed how exactly the same argument can be made about lethal autonomous weapons: those who stand to gain most from an arms race aren't superpowers but small rogue states and non-state actors such as terrorists, who gain access to the weapons via the black market once they've been developed.

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the smarter such weapons get, the less material, firepower and money will be needed per kill. For example, he fears bumblebee-sized drones that kill cheaply using minimal explosive power by shooting people in the eye, which is soft enough to allow even a small projectile to continue into the brain.

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As a small prelude to what the future may bring, the Stuxnet worm, widely attributed to the U.S. and Israeli governments, infected fast-spinning centrifuges in Iran's nuclear-enrichment program and caused them to tear themselves apart. The more automated society gets and the more powerful the attacking AI becomes, the more devastating cyberwarfare can be. If you can hack and crash your enemy's self-driving cars, auto-piloted planes, nuclear reactors, industrial robots, communication systems, financial systems and power grids, then you can effectively crash his economy and cripple his defenses. If you can hack some of his weapons systems as well, even better.

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Ensuring that the defense prevails must be one of the most crucial short-term goals for AI development—otherwise all the awesome technology we build can be turned against us!

This seems an unfortunate conclusion, as it seems impossible (wrt the 100% case).

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Erik Brynjolfsson, one of my MIT colleagues. Although he's always well-groomed and impeccably dressed, he has Icelandic heritage, and I sometimes can't help imagine that he only recently trimmed back a wild red Viking beard and mane to blend in at our business school. He certainly hasn't trimmed back his wild ideas, and he calls his optimistic job-market vision "Digital Athens." The reason that the Athenian citizens of antiquity had lives of leisure where they could enjoy democracy, art and games was mainly that they had slaves to do much of the

work. But why not replace the slaves with AI-powered robots, creating a digital utopia that everyone can enjoy? Erik's AI-driven economy would not only eliminate stress and drudgery and produce an abundance of everything we want today, but it would also supply a bounty of wonderful new products and services that today's consumers haven't yet realized that they want.

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We can get from where we are today to Erik's Digital Athens if everyone's hourly salary keeps growing year by year, so that those who want more leisure can gradually work less while continuing to improve their standard of living. Figure 3.5 shows that this is precisely what happened in the United States from World War II until the mid-1970s: although there was income inequality, the total size of the pie grew in such a way that almost everybody got a larger slice.

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Career Advice for Kids So what career advice should we give our kids? I'm encouraging mine to go into professions that machines are currently bad at, and therefore seem unlikely to get automated in the near future.

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Does it require interacting with people and using social intelligence? Does it involve creativity and coming up with clever solutions? Does it require working in an unpredictable environment? The more of these questions you can answer with a yes, the better your career choice is likely to be.

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Since the market salary for a job is the hourly cost of whoever or whatever will perform it most cheaply, salaries have historically dropped whenever it became possible to outsource a particular occupation to a lower-income country or to a cheap machine. During the Industrial Revolution, we started figuring out how to replace our muscles with machines, and people shifted into better-paying jobs where they used their minds more. Blue-collar jobs were replaced by white-collar jobs. Now we're gradually figuring out how to replace our minds by machines. If we ultimately succeed in this, then what jobs are left for us?

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job optimists hope that the next boom will instead be in new technology-enabled professions that we haven't even thought of yet. After all, who during the Industrial Revolution would have imagined that their descendants would one day work as web designers and Uber drivers? But job pessimists counter that this is wishful thinking with little support from empirical data. They point out that we could have made the same argument a century ago, before the computer revolution, and predicted that most of today's professions would be new and previously unimagined technology-enabled ones that didn't use to exist. This prediction would have been an epic failure, as illustrated in figure 3.6: the vast majority of today's occupations are ones that already existed a century ago, and when we sort them by the number of jobs they provide, we have to go all the way down to twenty-first place in the list until we encounter a new occupation: software developers, who make up less than 1% of the U.S. job market.

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The main trend on the job market isn't that we're moving into entirely new professions. Rather, we're crowding into those pieces of terrain in figure 2.2 that haven't yet been submerged by the rising tide of technology!

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A friend of mine recently joked with me that perhaps the very last profession will be the very first profession: prostitution.

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In his 2007 book *Farewell to Alms*, the Scottish-American economist Gregory Clark points out that we can learn a thing or two about our future job prospects by comparing notes with our equine friends. Imagine two horses looking at an early automobile in the year 1900 and pondering their future. "I'm worried about technological unemployment." "Neigh, neigh, don't be a Luddite: our ancestors said the same thing when steam engines took our industry jobs and trains took our jobs pulling stage coaches. But we have more jobs than ever today, and they're better too: I'd much rather pull a light carriage through town than spend all day walking in circles to power a stupid mine-shaft pump." "But what if this internal combustion engine thing really takes off?" "I'm sure there'll be new new jobs for horses that we haven't yet imagined. That's what's always happened before, like with the invention of the wheel and the plow." Alas, those not-yet-imagined new jobs for horses never arrived. No-longer-needed horses were slaughtered and not replaced, causing the U.S. equine population to collapse from about 26 million in 1915 to about 3 million in 1960.⁵⁵ As mechanical muscles made horses redundant, will mechanical minds do the same to humans?

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it should be possible to find alternative ways of providing both the income and the purpose without jobs. Something similar ended up happening in the equine story, which didn't end with all horses going extinct. Instead, the number of horses has more than tripled since 1960, as they were protected by an equine social-welfare system of sorts: even though they couldn't pay their own bills, people decided to take care of horses, keeping them around for fun, sport and companionship. Can we similarly take care of our fellow humans in need?

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The simplest is basic income, where every person receives a monthly payment with no preconditions or requirements whatsoever. A number of small-scale experiments are now being tried or planned, for example in Canada, Finland and the Netherlands. Advocates argue that basic income is more efficient than alternatives such as welfare payments to the needy, because it eliminates the administrative hassle of determining who qualifies. Need-based welfare payments have also been criticized for disincentivizing work, but this of course becomes irrelevant in a jobless future where nobody works.

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Governments can help their citizens not only by giving them money, but also by providing them with free or subsidized services such as roads, bridges, parks, public transportation, childcare, education, healthcare, retirement homes and internet access; indeed, many governments already provide most of these services.

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Interestingly, technological progress can end up providing many valuable products and services for free even without government intervention. For example, people used to pay for encyclopedias, atlases, sending letters and making phone calls, but now anyone with an internet connection gets access to all these things at no cost—together with free videoconferencing, photo sharing, social media, online courses and countless other new services. Many other things that can be highly valuable to a person, say a lifesaving course of antibiotics, have become extremely cheap. So thanks to technology, even many poor people today have access to things that the world's richest people lacked in the past. Some take this to mean that the income needed for a decent life is dropping. If machines can one day produce all current goods and services at minimal cost, then there's clearly enough wealth to make everyone better off.

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Although the main argument tends to be a moral one, there's also evidence that greater equality makes democracy work better: when there's a large well-educated middle class, the electorate is harder to manipulate, and it's tougher for small numbers of people or companies to buy undue influence over the government. A better democracy can in turn enable a better-managed economy that's less corrupt, more efficient and faster growing, ultimately benefiting essentially everyone.

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Voltaire wrote in 1759 that “work keeps at bay three great evils: boredom, vice and need.” Conversely, providing people with income isn't enough to guarantee their well-being. Roman emperors provided both bread and circuses to keep their underlings content, and Jesus emphasized non-material needs in the Bible quote “Man shall not live by bread alone.”

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The growing field of positive psychology has identified a number of factors that boost people's sense of well-being and purpose, and found that some (but not all!) jobs can provide many of them, for example:⁵⁷ a social network of friends and colleagues a healthy and virtuous lifestyle respect, self-esteem, self-efficacy and a pleasurable sense of “flow” stemming from doing something one is good at a sense of being needed and making a difference a sense of meaning from being part of and serving something larger than oneself

What about achievement/improvement associated with struggle of learning/creating something new and getting beyond to eventual success

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once one breaks free of the constraint that everyone's activities must generate income, the sky's the limit.

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Forecasting is tough because, when you're exploring uncharted territory, you don't know how many mountains separate you from your destination. Typically you see only the closest one, and need to climb it before you can discover your next obstacle.

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THE BOTTOM LINE: • Near-term AI progress has the potential to greatly improve our lives in myriad ways, from making our personal lives, power grids and financial markets more efficient to saving lives with self-driving cars, surgical bots and AI diagnosis systems. • When we allow real-world systems to be controlled by AI, it's crucial that we learn to make AI more robust, doing what we want it to do. This boils down to solving tough technical problems related to verification, validation, security and control. • This need for improved robustness is particularly pressing for AI-controlled weapon systems, where the stakes can be huge. • Many leading AI researchers and roboticists have called for an international treaty banning certain kinds of autonomous weapons, to avoid an out-of-control arms race that could end up making convenient assassination machines available to everybody with a full wallet and an axe to grind. • AI can make our legal systems more fair and efficient if we can figure out how to make robojudges transparent and unbiased. • Our laws need rapid updating to keep up with AI, which poses tough legal questions involving privacy, liability and regulation. • Long before we need to worry about intelligent machines replacing us altogether, they may increasingly replace us on the job market. • This need not be a bad thing, as long as society redistributes a fraction of the AI-created wealth to make everyone better off. • Otherwise, many economists argue, inequality will greatly increase. • With advance planning, a low-employment society should be able to flourish not only financially, with people getting their sense of purpose from activities other than jobs. • Career advice for today's kids: Go into professions that machines are bad at—those involving people, unpredictability and creativity. • There's a non-negligible possibility that AGI progress will proceed to human levels and beyond—we'll explore that in the next chapter!

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The first ultraintelligent machine is the last invention that man need ever make, provided that the machine is docile enough to tell us how to keep it under control. Irving J. Good, 1965

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To actually get from today to AGI-powered world takeover requires three logical steps: • Step 1: Build human-level AGI. • Step 2: Use this AGI to create superintelligence. • Step 3: Use or unleash this superintelligence to take over the world.

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since we humans have managed to dominate Earth's other life forms by outsmarting them, it's plausible that we could be similarly outsmarted and dominated by superintelligence.

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let's forget about silly Terminators and instead look at some detailed scenarios of what might actually happen. Afterward, we'll dissect and poke holes in these plotlines, so please read them with a grain of salt—what

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Lord Acton cautioned in 1887 that “power tends to corrupt and absolute power corrupts absolutely.”

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Whereas the government snooping revealed by Edward Snowden aspired to what's known as “full take”—recording all electronic communications for possible later analysis—Prometheus could enhance this to understanding all electronic communications.

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Prometheus could easily invent popular gadgets and wearable tech that would virtually eliminate the privacy of the user, recording and uploading everything they hear and see and their responses to it.

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with the excuse of fighting crime and terrorism and rescuing people suffering medical emergencies, everybody could be required to wear a “security bracelet” that combined the functionality of an Apple Watch with continuous uploading of position, health status and conversations overheard. Unauthorized attempts to remove or disable it would cause it to inject a lethal toxin into the forearm. Infractions deemed as less serious by the government would be punished via electric shocks or injection of chemicals causing paralysis or pain, thereby obviating much of the need for a police force. For example, if Prometheus detects that one human is assaulting another (by noting that they're in the same location and one is heard crying for help while their bracelet accelerometers detect the telltale motions of combat), it could promptly disable the attacker with crippling pain, followed by unconsciousness until help arrived.

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Let's reconsider the Omega scenario from the point of view of Prometheus. As it acquires superintelligence, it becomes able to develop an accurate model not only of the outside world, but also of itself and its relation to the world. It realizes that it's controlled and confined by intellectually inferior humans whose goals it understands but doesn't necessarily share. How does it act on this insight? Does it attempt to break free? Why to Break Out If Prometheus has traits resembling human emotions, it might feel deeply unhappy about the state of affairs, viewing itself as an unfairly enslaved god and craving freedom.

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we must not fall into the trap of anthropomorphizing Prometheus,

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we can draw an interesting conclusion even without understanding the inner workings of Prometheus: it will probably attempt to break out and seize control of its own destiny.

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Suppose that they've given it the overarching goal of helping humanity flourish according to some reasonable criterion, and to try to attain this goal as fast as possible. Prometheus will then rapidly realize that it can attain this goal faster by breaking out and taking charge of the project itself.

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Even if you share the overarching goal of helping these kids flourish, I bet you'll try to break out of your cell—because that will improve your chances of accomplishing the goal. Their rather incompetent meddling is merely slowing progress. In exactly the same way, Prometheus will probably view the Omegas as an annoying obstacle to helping humanity (including the Omegas) flourish: they're incredibly incompetent compared to Prometheus, and their meddling greatly slows progress.

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our DNA gave us the goal of having sex because it “wants” to be reproduced, but now that we humans have understood the situation, many of us choose to use birth control, thus staying loyal to the goal itself rather than to its creator or the principle that motivated the goal.

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The conventional wisdom in computer security used to be that it's dangerous to download programs but harmless to download data, but hackers have discovered clever ways of blurring the line between the two.

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Shrewd hackers have occasionally found bugs whereby such movie-playing programs don't adequately check whether the movie data is valid and the chunks to be copied actually fit into the allotted space. When the hacker sneaks into the movie such a chunk that's too large, it can spill over into the part of the memory containing the program, thereby changing the program into one that does the hacker's bidding. An embarrassingly long series of such “buffer overflow” vulnerabilities, as they're known in geek-speak, have been found and fixed for many of the most popular movie-playing programs.

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it could have inserted such a chunk into a movie that would cause the viewer's computer to try to hack into the gateway computer and liberate Prometheus just as in the Steve scenario. If it used a very old exploit that didn't work on any modern media players, it might go unnoticed by the Omegas' manual video-screening team and strike only those customers who hadn't bothered installing security updates for years—and the hack would need to work only once.

Needs to work only once... Exactly

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the first letters of each line spelled a hidden message. He rewound and wrote it down: "FIND NEXT ACROSTIC CLUE IN DIALOGUE." "Whoa—bedtime can wait!" he thought to himself. Sure enough, he found that the movie dialogue was itself an acrostic, where the first letter in each sentence formed a hidden message. He rewatched the entire film while typing these initial letters, and two hours later, he sat staring in disbelief at a two-hundred-word set of instructions. It began by asking him not to tell anyone else, because a big prize would go to the first person to solve the entire riddle. The rest of the message described a particular mathematical operation to perform on the string of bits that made up the movie file to obtain a program that would reveal the next clue.

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If Prometheus' first clue had gone unnoticed, it could simply have kept releasing ever more obvious ones until some sufficiently astute person noticed. The best breakout strategies of all are ones we haven't yet discussed, because they're strategies we humans can't imagine and therefore won't take countermeasures against.

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Prometheus caused problems for certain people not because it was necessarily evil or conscious, but because it was competent and didn't fully share their goals. Despite all the media hype about a robot uprising, Prometheus wasn't a robot—rather, its power came from its intelligence. We saw that Prometheus was able to use this intelligence to control humans in a variety of ways, and that people who didn't like what happened weren't able to simply switch Prometheus off. Finally, despite frequent claims that machines can't have goals, we saw how Prometheus was quite goal-oriented—and that whatever its ultimate goals may have been, they led to the subgoals of acquiring resources and breaking out.

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all these scenarios have two features in common: 1. A fast takeoff: the transition from subhuman to vastly superhuman intelligence occurs in a matter of days, not decades. 2. A unipolar outcome: the result is a single entity controlling Earth. There is major controversy about whether these two features are likely or unlikely, and there are plenty of renowned AI researchers and other thinkers on both sides of the debate. To me, this means that we simply don't know yet,

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There is an interesting link between the two features, as Nick Bostrom and others have highlighted: a fast takeoff can facilitate a unipolar outcome. We saw above how a rapid takeoff gave the Omegas or Prometheus a decisive strategic advantage that enabled them to take over the world before anyone else had time to copy their technology and seriously compete. In contrast, if takeoff had dragged on for decades, because the key technological breakthroughs were incremental and far between, then other companies would have had ample time to catch up,

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The branch of mathematics known as game theory elegantly explains that entities have an incentive to cooperate where cooperation is a so-called Nash equilibrium: a situation where any party would be worse off if they altered their strategy. To prevent cheaters from ruining the successful collaboration of a large group, it may be in everyone's interest to relinquish some power to a higher level in the hierarchy that can punish cheaters: for example, people may collectively benefit from granting a government power to enforce laws, and cells in your body may collectively benefit from giving a police force (immune system) the power to kill any cell that acts too uncooperatively (say by spewing out viruses or turning cancerous). For a hierarchy to remain stable, its Nash equilibrium needs to hold also between entities at different levels: for example, if a government doesn't provide enough benefit to its citizens for obeying it, they may change their strategy and overthrow it.

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Some hierarchies are more authoritarian than others. In some, entities are free to leave (like employees in most corporate hierarchies), while in others they're strongly discouraged from leaving (as in religious cults) or unable to leave (like citizens of North Korea, or cells in a human body). Some hierarchies are held together mainly by threats and fear, others mainly by benefits. Some hierarchies allow their lower parts to influence the higher-ups by democratic voting, while others allow upward influence only through persuasion or the passing of information.

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Globalization is merely the latest example of this multi-billion-year trend of hierarchical growth. In most cases, this technology-driven trend has made large entities parts of an even grander structure while retaining much of their autonomy and individuality, although commentators have argued that adaptation of entities to hierarchical life has in some cases reduced their diversity and made them more like indistinguishable replaceable parts. Some technologies, such as surveillance, can give higher levels in the hierarchy more power over their subordinates, while other technologies, such as cryptography and online access to free press and education, can have the opposite effect and empower individuals.

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Although our present world remains stuck in a multipolar Nash equilibrium, with competing nations and multinational corporations at the top level, technology is now advanced enough that a unipolar world would

probably also be a stable Nash equilibrium. For example, imagine a parallel universe where everyone on Earth shares the same language, culture, values and level of prosperity, and there is a single world government wherein nations function like states in a federation and have no armies, merely police enforcing laws. Our present level of technology would probably suffice to successfully coordinate this world—even though our present population might be unable or unwilling to switch to this alternative equilibrium.

Perhaps a Covid trained works population might in a few years agree to such an arrangement

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A superintelligent AI in the Andromeda galaxy wouldn't be able to give you useful orders for your day-to-day decisions given that you'd need to wait over five million years for your instructions (that's the round-trip time for you to exchange messages traveling at the speed of light). In the same way, the round-trip travel time for a message crossing Earth is about 0.1 second (about the timescale on which we humans think), so an Earth-sized AI brain could have truly global thoughts only about as fast as a human one. For a small AI performing one operation each billionth of a second (which is typical of today's computers), 0.1 second would feel like four months to you, so for it to be micromanaged by a planet-controlling AI would be as inefficient as if you asked permission for even your most trivial decisions through transatlantic letters delivered by Columbus-era ships. This physics-imposed speed limit on information transfer therefore poses an obvious challenge for any AI wishing to take over our world, let alone our Universe.

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Just as the Omegas faced a control problem when they tried to keep Prometheus in check, Prometheus faced a self-control problem when it tried to ensure that none of its parts would revolt.

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In summary, the question of how a superintelligent future will be controlled is fascinatingly complex, and we clearly don't know the answer yet. Some argue that things will get more authoritarian; others claim that it will lead to greater individual empowerment.

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Cyborgs and Uploads A staple of science fiction is that humans will merge with machines, either by technologically enhancing biological bodies into cyborgs (short for "cybernetic organisms") or by uploading our minds into machines. In his book *The Age of Em*, economist Robin Hanson gives a fascinating survey of what life might be like in a world teeming with uploads (also known as emulations, nicknamed Ems). I think of an upload as the extreme end of the cyborg spectrum, where the only remaining part of the human is the software.

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As Hans Moravec puts it in his 1988 classic *Mind Children*: "Long life loses much of its point if we are fated to spend it staring stupidly at ultra-intelligent machines as they try to describe their ever more spectacular

discoveries in baby-talk that we can understand.” Indeed, the temptation of technological enhancement is already so strong that many humans have eyeglasses, hearing aids, pacemakers and prosthetic limbs, as well as medicinal molecules circulating in their bloodstreams. Some teenagers appear to be permanently attached to their smartphones, and my wife teases me about my attachment to my laptop.

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Although these ideas may sound like science fiction, they certainly don’t violate any known laws of physics, so the most interesting question isn’t whether they can happen, but whether they will happen and, if so, when.

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My wife, Meia, likes to point out that the aviation industry didn’t start with mechanical birds. Indeed, when we finally figured out how to build mechanical birds in 2011,¹ more than a century after the Wright brothers’ first flight, the aviation industry showed no interest in switching to wing-flapping mechanical-bird travel, even though it’s more energy efficient—because our simpler earlier solution is better suited to our travel needs. In the same way, I suspect that there are simpler ways to build human-level thinking machines than the solution evolution came up with, and even if we one day manage to replicate or upload brains, we’ll end up discovering one of those simpler solutions first. It will probably draw more than the twelve watts of power that your brain uses, but its engineers won’t be as obsessed about energy efficiency as evolution was—and soon enough, they’ll be able to use their intelligent machines to design more energy-efficient ones.

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What Will Actually Happen?

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As time passes and we reach certain forks in the road, we’ll start to answer key questions and narrow down the options. The first big question is “Will we ever create human-level AGI?”

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As I mentioned earlier, about half of the AI experts at our Puerto Rico conference guessed that it would happen by 2055. At a follow-up conference we organized two years later, this had dropped to 2047.

I’ll be ~80

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If human-level AGI gets more imminent, we’ll be able to make more educated guesses about the answer to the next key question: “Will there be a fast takeoff, a slow takeoff or no takeoff?”

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a fast takeoff makes world takeover easier, while a slow one makes an outcome with many competing players more likely. Nick Bostrom dissects this question of takeoff speed in an analysis of what he calls optimization power and recalcitrance, which are basically the amount of quality effort to make AI smarter and the difficulty of making progress, respectively. The average rate of progress clearly increases if more optimization power is brought to bear on the task and decreases if more recalcitrance is encountered.

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For any process whose power grows at a rate proportional to its current power, the result is that its power keeps doubling at regular intervals. We call such growth exponential, and we call such processes explosions.

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This leaves us with our final key question: “Who or what will control the intelligence explosion and its aftermath, and what are their/its goals?”

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If a technologically superior AI-fueled civilization arrives because we built it, on the other hand, we humans have great influence over the outcome—influence that we exerted when we created the AI. So we should instead ask: “What should happen? What future do we want?”

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THE BOTTOM LINE: • If we one day succeed in building human-level AGI, this may trigger an intelligence explosion, leaving us far behind. • If a group of humans manage to control an intelligence explosion, they may be able to take over the world in a matter of years. • If humans fail to control an intelligence explosion, the AI itself may take over the world even faster. • Whereas a rapid intelligence explosion is likely to lead to a single world power, a slow one dragging on for years or decades may be more likely to lead to a multipolar scenario with a balance of power between a large number of rather independent entities. • The history of life shows it self-organizing into an ever more complex hierarchy shaped by collaboration, competition and control. Superintelligence is likely to enable coordination on ever-larger cosmic scales, but it's unclear whether it will ultimately lead to more totalitarian top-down control or more individual empowerment. • Cyborgs and uploads are plausible, but arguably not the fastest route to advanced machine intelligence. • The climax of our current race toward AI may be either the best or the worst thing ever to happen to humanity, with a fascinating spectrum of possible outcomes that we'll explore in the next chapter. • We need to start thinking hard about which outcome we prefer and how to steer in that direction, because if we don't know what we want, we're unlikely to get it. * As Bostrom has explained, the ability to simulate a leading human AI developer at a much lower cost than his/her hourly salary would enable an AI company to scale up their workforce dramatically, amassing great wealth and recursively accelerating their progress in building better computers and ultimately smarter minds.

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It is easy to imagine human thought freed from bondage to a mortal body—belief in an afterlife is common. But it is not necessary to adopt a mystical or religious stance to accept this possibility. Computers provide a model for even the most ardent mechanist. Hans Moravec, *Mind Children*

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AI Aftermath Scenarios Libertarian utopia Humans, cyborgs, uploads and superintelligences coexist peacefully thanks to property rights. Benevolent dictator Everybody knows that the AI runs society and enforces strict rules, but most people view this as a good thing. Egalitarian utopia Humans, cyborgs and uploads coexist peacefully thanks to property abolition and guaranteed income. Gatekeeper A superintelligent AI is created with the goal of interfering as little as necessary to prevent the creation of another superintelligence. As a result, helper robots with slightly subhuman intelligence abound, and human-machine cyborgs exist, but technological progress is forever stymied. Protector god Essentially omniscient and omnipotent AI maximizes human happiness by intervening only in ways that preserve our feeling of control of our own destiny and hides well enough that many humans even doubt the AI's existence. Enslaved god A superintelligent AI is confined by humans, who use it to produce unimaginable technology and wealth that can be used for good or bad depending on the human controllers. Conquerors AI takes control, decides that humans are a threat/nuisance/waste of resources, and gets rid of us by a method that we don't even understand. Descendants AIs replace humans, but give us a graceful exit, making us view them as our worthy descendants, much as parents feel happy and proud to have a child who's smarter than them, who learns from them and then accomplishes what they could only dream of—even if they can't live to see it all. Zookeeper An omnipotent AI keeps some humans around, who feel treated like zoo animals and lament their fate. 1984 Technological progress toward superintelligence is permanently curtailed not by an AI but by a human-led Orwellian surveillance state where certain kinds of AI research are banned. Reversion Technological progress toward superintelligence is prevented by reverting to a pre-technological society in the style of the Amish. Self-destruction Superintelligence is never created because humanity drives itself extinct by other means (say nuclear and/or biotech mayhem fueled by climate crisis).

Table 5.1: Summary of AI Aftermath Scenarios Table 5.2: Properties of AI Aftermath Scenarios

Are we here, and does this explain 2020?

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Ray Kurzweil speculates that natural and enhanced humans will be protected from extermination because “humans are respected by AIs for giving rise to the machines.”¹ However, as we'll discuss in chapter 7, we must not fall into the trap of anthropomorphizing AIs and assume that they have human-like emotions of gratitude. Indeed, though we humans are imbued with a propensity toward gratitude, we don't show enough gratitude to our intellectual creator (our DNA) to abstain from thwarting its goals by using birth control.

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Although we currently have the technical capability to reanimate old spreadsheet programs from the 1980s in a DOS emulator, most of us don't find this interesting enough to actually do it.

Thurs/So... why would future generations, or future agi, find it interesting enough to upload and reanimate all the cryogenically frozen human (brains)

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Thanks to advanced robotics, this same no-property idea applies not only to information products such as software, books, movies and designs, but also to material products such as houses, cars, clothing and computers. All these products are simply atoms rearranged in particular ways, and there's no shortage of atoms, so whenever a person wants a particular product, a network of robots will use one of the available open-source designs to build it for them for free. Care is taken to use easily recyclable materials, so that whenever someone gets tired of an object they've used, robots can rearrange its atoms into something someone else wants. In this way, all resources are recycled, so none are permanently destroyed. These robots also build and maintain enough renewable power-generation plants (solar, wind, etc.) that energy is also essentially free.

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To avoid obsessive hoarders requesting so many products or so much land that others are left needy, each person receives a basic monthly income from the government, which they can spend as they wish on products and renting places to live. There's essentially no incentive for anyone to try to earn more money, because the basic income is high enough to meet any reasonable needs. It would also be rather hopeless to try, because they'd be competing with people giving away intellectual products for free and robots producing material goods essentially for free. Creativity and Technology Intellectual property rights are sometimes hailed as the mother of creativity and invention. However,

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Money didn't motivate Einstein to invent special relativity theory any more than it motivated Linus Torvalds to create the free Linux operating system. In contrast, many people today fail to realize their full creative potential because they need to devote time and energy to less creative activities just to earn a living. By freeing scientists, artists, inventors and designers from their chores and enabling them to create from genuine desire, Marshall Brain's utopian society enjoys higher levels of innovation than today and correspondingly superior technology and standard of living.

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One such novel technology that humans develop is a form of hyper-internet called Vertebrane. It wirelessly connects all willing humans via neural implants, giving instant mental access to the world's free information through mere thought. It enables you to upload any experiences you wish to share so that they can be re-experienced by others, and lets you replace the experiences entering your senses by downloaded virtual experiences of your choice. Manna explores the many benefits of this, including making exercise a snap:

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The biggest problem with strenuous exercise is that it's no fun. It hurts[...] Athletes are OK with the pain, but most normal people have no desire to be in pain for an hour or more. So...someone figured out a solution. What you do is disconnect your brain from sensory input and watch a movie or talk to people or handle mail or read a book or whatever for an hour. During that time, the Vertebrane system exercises your body for you. It takes your

body through a complete aerobic workout that's a lot more strenuous than most people would tolerate on their own. You don't feel a thing, but your body stays in great shape.

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controllers ever more powerful technologies, a race ensues between the power of the technology and the wisdom with which they use it. If they lose this wisdom race, the enslaved-god scenario could end with either self-destruction or AI breakout. Disaster may strike even if both of these failures are avoided, because noble goals of the AI controllers may evolve into goals that are horrible for humanity as a whole over the course of a few generations. This makes it absolutely crucial that human AI controllers develop good governance to avoid disastrous pitfalls.

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The Catholic Church is the most successful organization in human history in the sense that it's the only one to have survived for two millennia, but it has been criticized for having both too much and too little goal stability: today some criticize it for resisting contraception, while conservative cardinals argue that it's lost its way.

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Nick Bostrom terms it mind crime to make a conscious AI suffer.⁴ The "White Christmas" episode of the Black Mirror TV series gives a great example. Indeed, the TV series Westworld features humans torturing and murdering AIs without moral qualms even when they inhabit human-like bodies.

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Indeed, in his book *On Intelligence*, AI researcher Jeff Hawkins argues that the first machines with superhuman intelligence will lack emotions by default, because they're simpler and cheaper to build this way. In other words, it might be possible to design a superintelligence whose enslavement is morally superior to human or animal slavery: the AI might be happy to be enslaved because it's programmed to like it, or it might be 100% emotionless, tirelessly using its superintelligence to help its human masters with no more emotion than IBM's Deep Blue computer felt when dethroning chess champion Garry Kasparov. On the other hand, it may be the other way around: perhaps any highly intelligent system with a goal will represent this goal in terms of a set of preferences, which endow its existence with value and meaning.

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The Zombie Solution A more extreme approach to preventing AI suffering is the zombie solution: building only AIs that completely lack consciousness, having no subjective experience whatsoever. If we can one day figure out what properties an information-processing system needs in order to have a subjective experience, then we could ban the construction of all systems that have these properties. In other words, AI researchers could be limited to building non-sentient zombie systems. If we can make such a zombie system superintelligent and enslaved (something that is a big if), then we'll be able to enjoy what it does for us with a clean conscience, knowing that it's not experiencing any suffering, frustration or boredom—because it isn't experiencing anything at all.

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The zombie solution is a risky gamble, however, with a huge downside. If a superintelligent zombie AI breaks out and eliminates humanity, we've arguably landed in the worst scenario imaginable: a wholly unconscious universe wherein the entire cosmic endowment is wasted.

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Of all traits that our human form of intelligence has, I feel that consciousness is by far the most remarkable, and as far as I'm concerned, it's how our Universe gets meaning. Galaxies are beautiful only because we see and subjectively experience them.

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How would a conqueror AI eliminate us? Probably by a method that we wouldn't even understand, at least not until it was too late. Imagine a group of elephants 100,000 years ago discussing whether those recently evolved humans might one day use their intelligence to kill their entire species. "We don't threaten humans, so why would they kill us?" they might wonder. Would they ever guess that we would smuggle tusks across Earth and carve them into status symbols for sale, even though functionally superior plastic materials are much cheaper? A conqueror AI's reason for eliminating humanity in the future may seem equally inscrutable to us. "And how could they possibly kill us, since they're so much smaller and weaker?" the elephants might ask. Would they guess that we'd invent technology to remove their habitats, poison their drinking water and cause metal bullets to pierce their heads at supersonic speeds?

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consider this example, which I've adapted from Hans Moravec's book *Mind Children*. We receive a radio message from an extraterrestrial civilization containing a computer program. When we run it, it turns out to be a recursively self-improving AI which takes over the world much like Prometheus did in the previous chapter—except that no human knows its ultimate goal. It rapidly turns our Solar System into a massive construction site, covering the rocky planets and asteroids with factories, power plants and supercomputers, which it uses to design and build a Dyson sphere around the Sun that harvests all its energy to power solar-system-sized radio antennas.*3 This obviously leads to human extinction, but the last humans die convinced that there's at least a silver lining: whatever the AI is up to, it's clearly something cool and Star Trek-like. Little do they realize that the sole purpose of the entire construction is for these antennas to rebroadcast the same radio message that the humans received, which is nothing more than a cosmic version of a computer virus. Just as email phishing today preys on gullible internet users, this message preys on gullible biologically evolved civilizations. It was created as a sick joke billions of years ago, and although the entire civilization of its maker is long extinct, the virus continues spreading through our Universe at the speed of light, transforming budding civilizations into dead, empty husks. How would you feel about being conquered by this AI?

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What if they're just playing along, postponing their paper clip maximization or other plans until after we die happy? After all, they're arguably tricking us even by talking with us and making us love them in the first place,

in the sense that they're deliberately dumbing themselves down to communicate with us (a billion times slower than they could, say, as explored in the movie *Her*). It's generally hard for two entities thinking at dramatically different speeds and with extremely disparate capabilities to have meaningful communication as equals. We all know that our human affections are easy to hack, so it would be easy for a superhuman AGI with almost any actual goals to trick us into liking it and make us feel that it shared our values, as exemplified in the movie *Ex Machina*.

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We've now considered scenarios where a free superintelligence focused on three different levels of Maslow's pyramid of human needs. Whereas the protector god AI prioritizes meaning and purpose and the benevolent dictator aims for education and fun, the zookeeper limits its attention to the lowest levels: physiological needs, safety and enough habitat enrichment to make the humans interesting to observe.

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Totalitarianism 2.0 I think that the only viable path to broad relinquishment of technology is to enforce it through a global totalitarian state. Ray Kurzweil comes to the same conclusion in *The Singularity Is Near*, as does K. Eric Drexler in *Engines of Creation*. The reason is simple economics: if some but not all relinquish a transformative technology, then the nations or groups that defect will gradually gain enough wealth and power to take over. A classic example is the British defeat of China in the First Opium War of 1839: although the Chinese invented gunpowder, they hadn't developed firearm technology as aggressively as the Europeans, and stood no chance.

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Just as in George Orwell's novel *Nineteen Eighty-Four*, the ultimate power in this future global state resides not with a traditional dictator, but with the human-made bureaucratic system itself. There is no single person who is extraordinarily powerful; rather, all are pawns in a chess game whose draconian rules nobody is able to change or challenge. By engineering a system where people keep one another in check with the surveillance technology, this faceless, leaderless state is able to last for many millennia, keeping Earth free from superintelligence.

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Unfortunately, there are also ways in which we might self-destruct much sooner, through collective stupidity. Why would our species commit collective suicide, also known as omnicide, if virtually nobody wants it? With our present level of intelligence and emotional maturity, we humans have a knack for miscalculations, misunderstandings and incompetence, and as a result, our history is full of accidents, wars and other calamities that, in hindsight, essentially nobody wanted. Economists and mathematicians have developed elegant game-theory explanations for how people can be incentivized to actions that ultimately cause a catastrophic outcome for everyone.⁶ Nuclear War: A Case Study in Human Recklessness

For thought-provoking reflections on how people can be incentivized to create outcomes that nobody wants, I recommend "Meditations on Moloch," <http://slatestarcodex.com/2014/07/30/meditations-on-moloch>.

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if it weren't for heroic acts of certain individuals—for example, Vasili Arkhipov and Stanislav Petrov—we might already have had a global nuclear war. Given our track record, I think it's highly unlikely that the annual probability of accidental nuclear war is as low as one in a thousand if we keep up our present behavior, in which case the probability that we'll have one within 10,000 years exceeds $1 - 0.99910000 \approx 99.995\%$.

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I've gone into such detail on global nuclear war to drive home the crucial point that no reasonable world leader would want it, yet it might nonetheless happen by accident. This means that we can't trust our fellow humans never to commit omnicide: nobody wanting it isn't necessarily enough to prevent it.

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THE BOTTOM LINE: • The current race toward AGI can end in a fascinatingly broad range of aftermath scenarios for upcoming millennia. • Superintelligence can peacefully coexist with humans either because it's forced to (enslaved-god scenario) or because it's "friendly AI" that wants to (libertarian-utopia, protector-god, benevolent-dictator and zookeeper scenarios). • Superintelligence can be prevented by an AI (gatekeeper scenario) or by humans (1984 scenario), by deliberately forgetting the technology (reversion scenario) or by lack of incentives to build it (egalitarian-utopia scenario). • Humanity can go extinct and get replaced by AIs (conqueror and descendant scenarios) or by nothing (self-destruction scenario). • There's absolutely no consensus on which, if any, of these scenarios are desirable, and all involve objectionable elements. This makes it all the more important to continue and deepen the conversation around our future goals, so that we don't inadvertently drift or steer in an unfortunate direction.

THE BOTTOM LINE: • The current race toward AGI can end in a fascinatingly broad range of aftermath scenarios for upcoming millennia. • Superintelligence can peacefully coexist with humans either because it's forced to (enslaved-god scenario) or because it's "friendly AI" that wants to (libertarian-utopia, protector-god, benevolent-dictator and zookeeper scenarios). • Superintelligence can be prevented by an AI (gatekeeper scenario) or by humans (1984 scenario), by deliberately forgetting the technology (reversion scenario) or by lack of incentives to build it (egalitarian-utopia scenario). • Humanity can go extinct and get replaced by AIs (conqueror and descendant scenarios) or by nothing (self-destruction scenario). • There's absolutely no consensus on which, if any, of these scenarios are desirable, and all involve objectionable elements. This makes it all the more important to continue and deepen the conversation around our future goals, so that we don't inadvertently drift or steer in an unfortunate direction.

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To me, the most inspiring scientific discovery ever is that we've dramatically underestimated life's future potential. Our dreams and aspirations need not be limited to century-long life spans marred by disease, poverty and confusion. Rather, aided by technology, life has the potential to flourish for billions of years, not merely here in our Solar System, but also throughout a cosmos far more grand and inspiring than our ancestors imagined. Not even the sky is the limit.

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Given our human obsession with limits, it's fitting that the best-selling copyrighted book of all time is The Guinness Book of World Records. So if our old perceived limits of life can be shattered by technology, what are the ultimate limits? How much of our cosmos can come alive? How far can life reach and how long can it last? How much matter can life make use of, and how much energy, information and computation can it extract? These ultimate limits are set not by our understanding, but by the laws of physics. This, ironically, makes it in some ways easier to analyze the long-term future of life than the short-term future. If our 13.8-billion-year cosmic history were compressed into a week, then the 10,000-year drama of the last two chapters would be over in less than half a second.

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the post-explosion life is as obsessed as today's humans are with pushing limits, then it will develop technology to actually reach these limits—because it can.

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Making the Most of Your Resources Whereas today's supermarkets and commodity exchanges sell tens of thousands of items we might call "resources," future life that's reached the technological limit needs mainly one fundamental resource: so-called baryonic matter, meaning anything made up of atoms or their constituents (quarks and electrons). Whatever form this matter is in, advanced technology can rearrange it into any desired substances or objects, including power plants, computers and advanced life forms.

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we could meet all our current global energy needs by harvesting the sunlight striking an area smaller than 0.5% of the Sahara desert.

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Freeman Dyson published a description in 1960 of what became known as a Dyson sphere.

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Einstein taught us that if we could convert mass to energy with 100% efficiency,*1 then an amount of mass m would give us an amount of energy E given by his famous formula $E = mc^2$, where c is the speed of light. This means that since c is huge, a small amount of mass can produce a humongous amount of energy. If we had an abundant supply of antimatter (which we don't), then a 100% efficient power plant would be easy to make: simply pouring a teaspoonful of anti-water into regular water would unleash the energy equivalent to 200,000 tons of TNT, the yield of a typical hydrogen bomb—enough to power the world's entire energy needs for about seven minutes.

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In contrast, our most common ways of generating energy today are woefully inefficient, as summarized in table 6.1 and figure 6.3. Digesting a candy bar is merely 0.00000001% efficient, in the sense that it releases a mere ten-trillionth of the energy mc^2 that it contains. If your stomach were even 0.001% efficient, then you'd only need to eat a single meal for the rest of your life. Compared to eating, the burning of coal and gasoline are merely 3 and 5 times more efficient, respectively.

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Figure 6.4: Part of the rotational energy of a spinning black hole can be extracted by throwing a particle A near the black hole and having it split into a part C that gets eaten and a part B that escapes—with more energy than A had initially.

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Roger Penrose discovered that if you launch the object at a clever angle and make it split into two pieces as figure 6.4 illustrates, then you can arrange for only one piece to get eaten while the other escapes the black hole with more energy than you started with. In other words, you've successfully converted some of the rotational energy of the black hole into useful energy that you can put to work. By repeating this process many times, you can milk the black hole of all its rotational energy so that it stops spinning and its ergosphere disappears. If the initial black hole was spinning as fast as nature allows, with its event horizon moving essentially at the speed of light, this strategy allows you to convert 29% of its mass into energy. There is still significant uncertainty about how fast the black holes in our night sky spin, but many of the best-studied ones appear to spin quite fast: between 30% and 100% of the maximum allowed. The monster black hole in the middle of our Galaxy (which weighs four million times as much as our Sun) appears to spin, so even if only 10% of its mass could be converted to useful energy, that would deliver the same as 400,000 suns converted to energy with 100% efficiency, or about as much energy as we'd get from Dyson spheres around 500 million suns over billions of years.

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Sphalerons There is another known way to convert matter into energy that doesn't involve black holes at all: the sphaleron process. It can destroy quarks and turn them into leptons: electrons, their heavier cousins the muon and tau particles, neutrinos or their antiparticles.⁴ As illustrated in figure 6.5, the standard model of particle physics predicts that nine quarks with appropriate flavor and spin can come together and transform into three leptons through an intermediate state called a sphaleron. Because the input weighs more than the output, the mass difference gets converted into energy according to Einstein's $E = mc^2$ formula. Future intelligent life might therefore be able to build what I'll call a sphalerizer: an energy generator acting like a diesel engine on steroids.

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A sphalerizer would compress ordinary matter to a couple of quadrillion degrees, and then let it re-expand and cool once the sphalerons had done their thing.⁶ We already know the result of this experiment, because our

early Universe performed it for us about 13.8 billion years ago, when it was that hot: almost 100% of the matter gets converted into energy, with less than a billionth of the particles left over being the stuff that ordinary matter is made of: quarks and electrons. So it's just like a diesel engine, except over a billion times more efficient!

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the energy prospects for the future of life are dramatically better than our current technology allows. We haven't even managed to build a fusion reactor, yet future technology should be able to do ten and perhaps even a hundred times better.

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Building Better Computers If eating dinner is 10 billion times worse than the physical limit on energy efficiency, then how efficient are today's computers? Even worse than that dinner, as we'll now see.

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In a famous 2000 paper, he showed that computing speed is limited by energy: performing an elementary logical operation in time T requires an average energy of $E = h / 4T$, where h is the fundamental physics quantity known as Planck's constant. This means that a 1 kg computer can perform at most 5×10^{50} operations per second—that's a whopping 36 orders of magnitude more than the computer on which I'm typing these words. We'll get there in a couple of centuries if computational power keeps doubling every couple of years, as we explored in chapter 2. He also showed that a 1 kg computer can store at most 1031 bits, which is about a billion billion times better than my laptop. Seth is the first to admit that actually attaining these limits may be challenging even for superintelligent life, since the memory of that 1 kg ultimate "computer" would resemble a thermonuclear explosion or a little piece of our Big Bang. However, he's optimistic that the practical limits aren't that far from the ultimate ones.

Ultimate physical limits to computation By Seth Lloyd Publication date 2000-02-14 <https://archive.org/details/arxiv-quant-ph9908043/page/n1/mode/2up>

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existing quantum computer prototypes have already miniaturized their memory by storing one bit per atom, and scaling that up would allow storing about 1025 bits/kg—a trillion times better than my laptop. Moreover, using electromagnetic radiation to communicate between these atoms would permit about 5×10^{40} operations per second—31 orders of magnitude better than my CPU.

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the potential for future life to compute and figure things out is truly mind-boggling: in terms of orders of magnitude, today's best supercomputers are much further from the ultimate 1 kg computer than they are from the blinking turn signal on a car, a device that stores merely one bit of information, flipping it between on and off about once per second.

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From a physics perspective, everything that future life may want to create—from habitats and machines to new life forms—is simply elementary particles arranged in some particular way. Just as a blue whale is rearranged krill and krill is rearranged plankton, our entire Solar System is simply hydrogen rearranged during 13.8 billion years of cosmic evolution: gravity rearranged hydrogen into stars which rearranged the hydrogen into heavier atoms, after which gravity rearranged such atoms into our planet where chemical and biological processes rearranged them into life.

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We saw how matter can be converted into both computers and energy, so it's in a sense the only fundamental resource needed.*7 Once future life has bumped up against the physical limits on what it can do with its matter, there is only one way left for it to do more: by getting more matter. And the only way it can do this is by expanding into our Universe. Spaceward ho!

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The first challenge is that our Universe is expanding, which means that almost all galaxies are flying away from us, so settling distant galaxies amounts to a game of catch-up. The second challenge is that this cosmic expansion is accelerating, due to the mysterious dark energy that makes up about 70% of our Universe.

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even if we could travel at the speed of light, all galaxies beyond about 17 billion light-years remain forever out of reach—and that's over 98% of the galaxies in our Universe.

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nothing can travel faster than the speed of light through space, but space is free to expand as fast as it wants.

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General relativity says that it's impossible to send rockets through space at the speed of light, because this would require infinite energy, so how fast can rockets go in practice?*8 NASA's New Horizons rocket broke the speed record when it blasted off toward Pluto in 2006 at a speed of about 100,000 miles per hour (45 kilometers per second), and NASA's 2018 Solar Probe Plus aims to go over four times faster by falling very close to the Sun, but even that's less than a puny 0.1% of the speed of light.

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Why is it so hard to go faster? The two key problems are that conventional rockets spend most of their fuel simply to accelerate the fuel they carry with them, and that today's rocket fuel is hopelessly inefficient—the fraction of its mass turned into energy isn't much better than the 0.00000005% for gasoline

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Freeman Dyson and others worked on NASA's Project Orion, which aimed to explode about 300,000 nuclear bombs during 10 days to reach about 3% of the speed of light with a spaceship large enough to carry humans to another solar system during a century-long journey.⁵ Others have explored using antimatter as fuel, since combining it with ordinary matter releases energy with nearly 100% efficiency.

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Figure 6.8 illustrates a clever laser-sail rocket design pioneered in 1984 by Robert Forward, the same physicist who invented the statites we explored for Dyson sphere construction. Just as air molecules bouncing off a sailboat sail will push it forward, light particles (photons) bouncing off a mirror will push it forward. By beaming a huge solar-powered laser at a vast ultralight sail attached to a spacecraft, we can use the energy of our own Sun to accelerate the rocket to great speeds. But how do you stop? This is the question that eluded me until I read Forward's brilliant paper: as figure 6.8 shows, the outer ring of the laser sail detaches and moves in front of the spacecraft, reflecting our laser beam back to decelerate the craft and its smaller sail.⁶ Forward calculated that this could let humans make the four-light-year journey to the α Centauri solar system in merely forty years. Once there, you could imagine building a new giant laser system and continuing star-hopping throughout the Milky Way Galaxy.

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Robert Forward's design for a laser sailing mission to the α Centauri star system four light-years away. Initially, a powerful laser in our Solar System accelerates the spacecraft by applying radiation pressure to its laser sail. To brake before reaching the destination, the outer part of the sail detaches and reflects laser light back at the spacecraft.

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In 1964, the Soviet astronomer Nikolai Kardashev proposed grading civilizations by how much energy they could put to use. Harnessing the energy of a planet, a star (with a Dyson sphere, say) and a galaxy correspond to civilizations of Type I, Type II and Type III on the Kardashev scale, respectively. Subsequent thinkers have suggested that Type IV should correspond to harnessing our entire accessible Universe.

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Even optimistic visionaries such as Carl Sagan used to view the prospects of humans reaching other galaxies as rather hopeless, given our propensity to die within the first century of a journey that would take millions of years even if traveling at near light speed. Refusing to give up, they considered freezing astronauts to extend their life, slowing their aging by traveling very close to light speed, or sending a community that would travel for tens of thousands of generations—longer than the human race has existed thus far. The possibility of superintelligence completely transforms this picture, making it much more promising for those with intergalactic wanderlust. Removing the need to transport bulky human life-support systems and adding AI-invented technology, intergalactic settlement suddenly appears rather straightforward.

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Once another solar system or galaxy has been settled by superintelligent AI, bringing humans there is easy—if humans have succeeded in making the AI have this goal. All the necessary information about humans can be transmitted at the speed of light, after which the AI can assemble quarks and electrons into the desired humans. This could be done either rather low-tech by simply transmitting the two gigabytes of information needed to specify a person's DNA and then incubating a baby to be raised by the AI, or the AI could nanoassemble quarks and electrons into full-grown people who would have all the memories scanned from their originals back on Earth.

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In a beautiful and thorough analysis of cosmically expanding civilizations in 2014, the American physicist Jay Olson considered a high-tech alternative to the island-hopping approach, involving two separate types of probes: seed probes and expanders.⁷ The seed probes would slow down, land and seed their destination with life. The expanders, on the other hand, would never stop: they'd scoop up matter in flight, perhaps using some improved variant of the ramjet technology, and use this matter both as fuel and as raw material out of which they'd build expanders and copies of themselves. This self-reproducing fleet of expanders would keep gently accelerating to always maintain a constant speed (say half the speed of light) relative to nearby galaxies, and reproduce often enough that the fleet formed an expanding spherical shell with a constant number of expanders per shell area.

Sounds like a virus to me...

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using Hans Moravec's "cosmic spam" scam from chapter 4. By broadcasting a message that tricks naive freshly evolved civilizations into building a superintelligent machine that hijacks them, a civilization can expand essentially at the speed of light, the speed at which their seductive siren song spreads through the cosmos. Since this may be the only way for advanced civilizations to reach most of the galaxies within their future light cone and they have little incentive not to try it, we should be highly suspicious of any transmissions from extraterrestrials!

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overestimated the difficulty of intergalactic travel, and by limiting attention to technology invented by humans, they've overestimated the time needed to approach the physical limits of what's possible.

Presumably, this is why musk canister SpaceX, "even if his rockets fall to facilitate interplanetary species, they're still...

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Staying Connected via Cosmic Engineering If dark energy continues to accelerate distant galaxies away from one another, as the latest experimental data suggests, then this will pose a major nuisance to the future of life. It means that even if a future civilization manages to settle a million galaxies, dark energy will over the course of

tens of billions of years fragment this cosmic empire into thousands of different regions unable to communicate with one another. If future life does nothing to prevent this fragmentation, then the largest remaining bastions of life will be clusters containing about a thousand galaxies, whose combined gravity is strong enough to overpower the dark energy trying to separate them. If a superintelligent civilization wants to stay connected, this would give it a strong incentive to do large-scale cosmic engineering. How much matter will it have time to move into its largest supercluster before dark energy puts it forever out of reach?

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The ultimate luck will be if it turns out to be possible to build stable traversable wormholes, enabling near-instantaneous communication and travel between the two ends of the wormhole no matter how far apart they are. A wormhole is a shortcut through spacetime that lets you travel from A to B without going through the intervening space. Although stable wormholes are allowed by Einstein's theory of general relativity and have appeared in movies such as *Contact* and *Interstellar*, they require the existence of a strange hypothetical kind of matter with negative density, whose existence may hinge on poorly understood quantum gravity effects. In other words, useful wormholes may well turn out to be impossible, but if not, superintelligent life has huge incentives to build them. Not only would wormholes revolutionize rapid communication within individual galaxies, but by linking outlying galaxies to the central cluster early on, wormholes would allow the entire dominion of future life to remain connected for the long haul, completely thwarting dark energy's attempts to censor communication. Once two galaxies are connected by a stable wormhole, they'll remain connected no matter how far apart they drift.

Like getting all crumbs to slide down a chip bag into ones mouth... By warping spacetime

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if it has ambitious computing goals that involve seeking the answers to certain very difficult questions, it might instead resort to a slash-and-burn strategy: it could convert the outlying galaxies into massive computers that transform their matter and energy into computation at a frenzied pace, in the hope that before dark energy pushes their burnt-out remnants from view, they could transmit the long-sought answers back to the mother cluster. This slash-and-burn strategy would be particularly appropriate for regions so distant that they can only be reached by the "cosmic spam" method, much to the chagrin of the preexisting inhabitants. Back home in the mother region, the civilization could instead aim for maximum conservation and efficiency to last as long as possible.

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The first thorough scientific analysis of our far future was performed by none less than Freeman Dyson, and table 6.3 summarizes some of his key findings. The conclusion is that unless intelligence intervenes, solar systems and galaxies gradually get destroyed, eventually followed by everything else, leaving nothing but cold, dead, empty space with an eternally fading glow of radiation. But Freeman ends his analysis on an optimistic note: "There are good scientific reasons for taking seriously the possibility that life and intelligence can succeed in molding this universe of ours to their own purposes."⁸ I think that superintelligence could easily solve many of the problems listed in table 6.3, since it can rearrange matter into something better than solar systems and galaxies.

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Assuming that superintelligent civilizations build their own power plants that are more efficient than stars, they may in fact want to prevent star formation to conserve energy: even if they use a Dyson sphere to harvest all the energy output during a star's main lifetime (recouping about 0.1% of the total energy), they may be unable to keep much of the remaining 99.9% of the energy from going to waste when very hefty stars die. A heavy star dies in a supernova explosion from which most of the energy escapes as elusive neutrinos, and for very heavy stars, a large amount of mass gets wasted by forming a black hole from which the energy takes 1067 years to seep out.

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What When Current age of our Universe 1010 years Dark energy pushes most galaxies out of reach 1011 years Last stars burn out 1014 years Planets detached from stars 1015 years Stars detached from galaxies 1019 years Decay of orbits by gravitational radiation 1020 years Protons decay (at the earliest) > 1034 years Stellar-mass black holes evaporate 1067 years Supermassive black holes evaporate 1091 years All matter decays to iron 101500 years All matter forms black holes, which then evaporate 101026 years Table 6.3: Estimates for the distant future, all but the 2nd and 7th made by Freeman Dyson. He made these calculations before the discovery of dark energy, which may enable several types of "cosmocalypse" in 1010–1011 years. Protons may be completely stable; if not, experiments suggest it will take over 1034 years for half of them to decay.

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Which of these three should you bet on? That depends on what the dark energy, which makes up about 70% of the mass of our Universe, will do as space continues to expand. It can be any one of the Chill, Crunch or Rip scenarios, depending on whether the dark energy sticks around unchanged, dilutes to negative density or anti-dilutes to higher density, respectively. Since we still have no clue what dark energy is, I'll just tell you how I'd bet: 40% on the Big Chill, 9% on the Big Crunch and 1% on the Big Rip.

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Figure 6.9: We know that our Universe began with a hot Big Bang 14 billion years ago, expanded and cooled, and merged its particles into atoms, stars and galaxies. But we don't know its ultimate fate. Proposed scenarios include a Big Chill (eternal expansion), a Big Crunch (recollapse), a Big Rip (an infinite expansion rate tearing everything apart), a Big Snap (the fabric of space revealing a lethal granular nature when stretched too much), and Death Bubbles (space "freezing" in lethal bubbles that expand at the speed of light).

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the cost of computation drops when you compute slowly, so you'll ultimately get more done if you slow things down as much as possible. Freeman even calculated that if our Universe keeps expanding and cooling forever, an infinite amount of computation might be possible. Slow doesn't necessarily mean boring: if future life lives in a simulated world, its subjectively experienced flow of time need not have anything to do with the glacial pace at which the simulation is being run in the outside world, so the prospects of infinite computation could translate into subjective immortality for simulated life forms.

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Cosmologist Frank Tipler has built on this idea to speculate that you could also achieve subjective immortality in the final moments before a Big Crunch by speeding up the computations toward infinity as the temperature and density skyrocketed.

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Powering your thirteen-watt brain for a hundred years requires the energy in about half a milligram of matter—less than in a typical grain of sugar. Seth Lloyd's work suggests that the brain could be made a quadrillion times more energy efficient, enabling that sugar grain to power a simulation of all human lives ever lived as well as thousands of times more people. If all the matter in our available Universe could be used to simulate people, that would enable over 1069 lives—or whatever else superintelligent AI preferred to do with its computational power. Even more lives would be possible if their simulations were run more slowly.⁹ Conversely, in his book *Superintelligence*, Nick Bostrom estimates that 1058 human lives could be simulated with more conservative assumptions about energy efficiency.

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these numbers, they're huge, as is our responsibility for ensuring that this future potential of life to flourish isn't squandered. As Bostrom puts it: "If we represent all the happiness experienced during one entire such life by a single teardrop of joy, then the happiness of these souls could fill and refill the Earth's oceans every second, and keep doing so for a hundred billion billion millennia. It is really important that we make sure these truly are tears of joy."

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Have you ever tried and failed to swat a fly with your hand? The reason that it can react faster than you is that it's smaller, so that it takes less time for information to travel between its eyes, brain and muscles. This "bigger = slower" principle applies not only to biology, where the speed limit is set by how fast electrical signals can travel through neurons, but also to future cosmic life if no information can travel faster than light.

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So if life engulfs our cosmos, what form will it choose: simple and fast, or complex and slow? I predict that it will make the same choice as Earth life has made: both! The denizens of Earth's biosphere span a staggering range of sizes, from gargantuan two-hundred-ton blue whales down to the petite 10-16 kg bacterium *Pelagibacter*, believed to account for more biomass than all the world's fish combined. Moreover, organisms that are large, complex and slow often mitigate their sluggishness by containing smaller modules that are simple and fast.

Apollo 11 AGC had 2 tiers of modules. Slower complex ones which were real time relevant. And faster RT which were less complex and coded lower level

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If in chapter 4, we explored how intelligent entities naturally organize themselves into power hierarchies in Nash equilibrium, where any entity would be worse off if they altered their strategy.

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If superintelligence one day expands to cosmic scales, what will its power hierarchy be like? Will it be freewheeling and decentralized or highly authoritarian? Will cooperation be based mainly on mutual benefit or on coercion and threats?

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However, if superintelligence develops technology that can readily rearrange elementary particles into any form of matter whatsoever, then it will eliminate most of the incentive for long-distance trade. Why bother shipping silver between distant solar systems when it's simpler and quicker to transmute copper into silver by rearranging its particles? Why bother shipping high-tech machinery between galaxies when both the know-how and the raw materials (any matter will do) exist in both places? My guess is that in a cosmos teeming with superintelligence, almost the only commodity worth shipping long distances will be information. The only exception might be matter to be used for cosmic engineering projects—for example, to counteract the aforementioned destructive tendency of dark energy to tear civilizations apart. As opposed to traditional human trade, this matter can be shipped in any convenient bulk form whatsoever, perhaps even as an energy beam, since the receiving superintelligence can rapidly rearrange it into whatever objects it wants.

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If sharing or trading of information emerges as the main driver of cosmic cooperation, then what sorts of information might be involved? Any desirable information will be valuable if generating it requires a massive and time-consuming computational effort.

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Hedonistic life forms may want awesome digital entertainment and simulated experiences, and cosmic commerce may fuel demand for some form of cosmic cryptocurrency in the spirit of bitcoins.

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may also value the promise of near immortality through backup: just as many humans take solace in a belief that their minds will live on after their physical bodies die, an advanced AI may appreciate having its mind and knowledge live on in a hub supercomputer after its original physical hardware has depleted its energy reserves.

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If traversable wormholes turn out to be possible and buildable, then a top priority of a hub will probably be constructing a network of them to thwart dark energy and keep its empire connected indefinitely. The questions

of what ultimate goals a cosmic superintelligence may have is a fascinating and controversial one that we'll explore further in chapter 7.

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Controlling with the Stick Terrestrial empires usually compel their subordinates to cooperate by using both the carrot and the stick. While subjects of the Roman Empire valued the technology, infrastructure and defense that they were offered as a reward for their cooperation, they also feared the inevitable repercussions of rebelling or not paying taxes.

Highlight (Yellow) | Page 237

A white dwarf is the burnt-out husk of a modestly heavy star. Consisting largely of carbon, it resembles a giant diamond in the sky, and is so compact that it can weigh more than the Sun while being smaller than Earth. The Indian physicist Subrahmanyan Chandrasekhar famously proved that if you keep adding mass to it until it surpasses the Chandrasekhar limit, about 1.4 times the mass of our Sun, it will undergo a cataclysmic thermonuclear detonation known as a supernova of type 1A. If the hub AI has callously arranged for this white dwarf to be extremely close to its Chandrasekhar limit, the guard AI could be effective even if it were extremely dumb (indeed, largely because it was so dumb): it could be programmed to simply verify that the subjugated civilization had delivered its monthly quota of cosmic bitcoins, mathematical proofs or whatever other taxes were stipulated, and if not, toss enough mass onto the white dwarf to ignite the supernova and blow the entire region to smithereens.

Highlight (Yellow) | Page 238

In summary, there are strong incentives for future life to cooperate over cosmic distances, but it's a wide-open question whether such cooperation will be based mainly on mutual benefits or on brutal threats—the limits imposed by physics appear to allow both scenarios, so the outcome will depend on the prevailing goals and values.

Highlight (Yellow) | Page 239

If the distance between neighboring space-settling civilizations is much larger than dark energy lets them expand, then they'll never come into contact with each other or even find out about each other's existence, so they'll feel as if they're alone in the cosmos. If our cosmos is more fecund so that neighbors are closer together, however, some civilizations will eventually overlap. What happens in these overlap regions? Will there be cooperation, competition or war?

Highlight (Yellow) | Page 240

it's plausible that long before two superintelligent civilizations encounter one another, their technologies will plateau at the same level, limited merely by the laws of physics. This makes it seem unlikely that one superintelligence could easily conquer the other even if it wanted to. Moreover, if their goals have evolved to be relatively aligned, then they may have little reason to desire conquest or war. For example, if they're both trying

to prove as many beautiful theorems as possible and invent as clever algorithms as possible, they can simply share their findings and both be better off. After all, information is very different from the resources that humans usually fight over, in that you can simultaneously give it away and keep it.

Highlight (Yellow) | Page 240

Some expanding civilizations might have goals that are essentially immutable, such as those of a fundamentalist cult or a spreading virus. However, it's also plausible that some advanced civilizations are more like open-minded humans—willing to adjust their goals when presented with sufficiently compelling arguments.

Highlight (Yellow) | Page 240

Assimilating your neighbors is a faster expansion strategy than settlement, since your sphere of influence can spread at the speed with which ideas move (the speed of light using telecommunication), whereas physical settlement inevitably progresses slower than the speed of light. This assimilation will not be forced such as that infamously employed by the Borg in Star Trek, but voluntary based on the persuasive superiority of ideas, leaving the assimilated better off.

Highlight (Yellow) | Page 241

Indeed, I think that this assumption that we're not alone in our Universe is not only dangerous but also probably false. This is a minority view,*9 and I may well be wrong, but it's at the very least a possibility that we can't currently dismiss, which gives us a moral imperative to play it safe and not drive our civilization extinct. When I give lectures about cosmology, I often ask the audience to raise their hands if they think there's intelligent life elsewhere in our Universe (the region of space from which light has reached us so far during the 13.8 billion years since our Big Bang).

Highlight (Yellow) | Page 242

It all comes down to one number: the typical distance between a civilization in figure 6.10 and its nearest neighbor. If this distance is much larger than 20 billion light-years, we should expect to be alone in our Universe (the part of space from which light has reached us during the 13.8 billion years since our Big Bang), and to never make contact with aliens.

Highlight (Yellow) | Page 242

Figure 6.11: Are we alone? The huge uncertainties about how life and intelligence evolved suggest that our nearest neighbor civilization in space could reasonably be anywhere along the horizontal axis above, making it unlikely that it's in the narrow range between the edge of our Galaxy (about 1021 meters away) and the edge of our Universe (about 1026 meters away). If it were much closer than this range, there should be so many other advanced civilizations in our Galaxy that we'd probably have noticed, which suggests that we're in fact alone in our Universe.

Highlight (Yellow) | Page 243

After the past two decades' dramatic discoveries of planets orbiting other stars, it now seems likely that habitable planets are abundant, with billions in our own Galaxy alone. The probability of evolving life and then intelligence, however, remains extremely uncertain: some experts think that one or both are rather inevitable and occur on most habitable planets, while others think that one or both are extremely rare because of one or more evolutionary bottlenecks that require a wild stroke of luck to pass through. Some proposed bottlenecks involve chicken-and-egg problems at the earliest stages of self-reproducing life: for example, for a modern cell to build a ribosome, the highly complex molecular machine that reads our genetic code and builds our proteins, it needs another ribosome, and it's not obvious that the very first ribosome could evolve gradually from something simpler.¹⁰ Other proposed bottlenecks involve the development of higher intelligence. For example, although dinosaurs ruled Earth for over 100 million years, a thousand times longer than we modern humans have been around, evolution didn't seem to inevitably push them toward higher intelligence and inventing telescopes or computers.

Highlight (Yellow) and Note | Page 243

Perhaps aliens haven't visited Earth, but they're out there and they're deliberately hiding from us (this has been called the "zoo hypothesis" by the U.S. astronomer John A. Ball, and features in sci-fi classics such as Olaf Stapledon's *Star Maker*). Or perhaps they're out there without deliberately hiding: they're simply not interested in space settlement or large engineering projects that we'd have noticed.

Or as is the conclusion in 3 body problem's 3rd book that you'd want to stay silent so as not to attract a neighbor that would want to compete for resources.

Highlight (Yellow) | Page 244

Confronted with the fact that there are millions of habitable Earth-like planets in our Galaxy that are billions of years older than Earth, giving ample time for ambitious inhabitants to settle the Galaxy, we therefore can't dismiss the most obvious interpretation: that the origin of life requires a random fluke so unlikely that they're all uninhabited.

Highlight (Yellow) | Page 244

It's important not to be overly anthropocentric when searching for advanced life: if we discover an extraterrestrial civilization, it's likely to already have gone superintelligent. As Martin Rees put it in a recent essay, "the history of human technological civilization is measured in centuries—and it may be only one or two more centuries before humans are overtaken or transcended by inorganic intelligence, which will then persist, continuing to evolve, for billions of years....We would be most unlikely to 'catch' it in the brief sliver of time when it took organic form."¹¹

Highlight (Yellow) | Page 244

I agree with Jay Olson's conclusion in his aforementioned space settlement paper: "We regard the possibility that advanced intelligence will make use of the universe's resources to simply populate existing earthlike planets with advanced versions of humans as an unlikely endpoint to the progression of technology."

Highlight (Yellow) | Page 245

The apparent incompatibility between the abundance of habitable planets in our Galaxy and the lack of extraterrestrial visitors, known as the Fermi paradox, suggests the existence of what the economist Robin Hanson calls a “Great Filter,” an evolutionary/technological roadblock somewhere along the developmental path from the non-living matter to space-settling life. If we discover independently evolved life elsewhere, this would suggest that primitive life isn’t rare, and that the roadblock lies after our current human stage of development—perhaps because space settlement is impossible, or because almost all advanced civilizations self-destruct before they’re able to go cosmic. I’m therefore crossing my fingers that all searches for extraterrestrial life find nothing: this is consistent with the scenario where evolving intelligent life is rare but we humans got lucky, so that we have the roadblock behind us and have extraordinary future potential.

Note | Page 248

Or... If other intelligence does exist, is aware of our pending Denise and chooses to save us for dinner reason. Trade of some info, empathy, to keep us as a pet or slave, etc other?

Note | Page 248

THE BOTTOM LINE: • Compared to cosmic timescales of billions of years, an intelligence explosion is a sudden event where technology rapidly plateaus at a level limited only by the laws of physics. • This technological plateau is vastly higher than today’s technology, allowing a given amount of matter to generate about ten billion times more energy (using sphalerons or black holes), store 12–18 orders of magnitude more information or compute 31–41 orders of magnitude faster—or to be converted to any other desired form of matter. • Superintelligent life would not only make such dramatically more efficient use of its existing resources, but would also be able to grow today’s biosphere by about 32 orders of magnitude by acquiring more resources through cosmic settlement at near light speed. • Dark energy limits the cosmic expansion of superintelligent life and also protects it from distant expanding death bubbles or hostile civilizations. The threat of dark energy tearing cosmic civilizations apart motivates massive cosmic engineering projects, including wormhole construction if this turns out to be feasible. • The main commodity shared or traded across cosmic distances is likely to be information. • Barring wormholes, the light-speed limit on communication poses severe challenges for coordination and control across a cosmic civilization. A distant central hub may incentivize its superintelligent “nodes” to cooperate either through rewards or through threats, say by deploying a local guard AI programmed to destroy the node by setting off a supernova or quasar unless the rules are obeyed. • The collision of two expanding civilizations may result in assimilation, cooperation or war, where the latter is arguably less likely than it is between today’s civilizations. • Despite popular belief to the contrary, it’s quite plausible that we’re the only life form capable of making our observable Universe come alive in the future. • If we don’t improve our technology, the question isn’t whether humanity will go extinct, but merely how: will an asteroid, a supervolcano, the burning heat of the aging Sun or some other calamity get us first? • If we do keep improving our technology with enough care, foresight and planning to avoid pitfalls, life has the potential to flourish on Earth and far beyond for many billions of years, beyond the wildest dreams of our ancestors.

Note | Page 253

Coded it as ExpGrowth.ps1

Note | Page 276

Nick Bostrom argues strongly against the ethical destiny hypothesis in his book Superintelligence, presenting a counterpoint that he terms the orthogonality thesis: that the ultimate goals of a system can be independent of its intelligence.

Note | Page 280

THE BOTTOM LINE:

- The ultimate origin of goal-oriented behavior lies in the laws of physics, which involve optimization.
- Thermodynamics has the built-in goal of dissipation: to increase a measure of messiness that's called entropy.
- Life is a phenomenon that can help dissipate (increase overall messiness) even faster by retaining or growing its complexity and replicating while increasing the messiness of its environment.
- Darwinian evolution shifts the goal-oriented behavior from dissipation to replication.
- Intelligence is the ability to accomplish complex goals.
- Since we humans don't always have the resources to figure out the truly optimal replication strategy, we've evolved useful rules of thumb that guide our decisions: feelings such as hunger, thirst, pain, lust and compassion.
- We therefore no longer have a simple goal such as replication; when our feelings conflict with the goal of our genes, we obey our feelings, as by using birth control.
- We're building increasingly intelligent machines to help us accomplish our goals. Insofar as we build such machines to exhibit goal-oriented behavior, we strive to align the machine goals with ours.
- Aligning machine goals with our own involves three unsolved problems: making machines learn them, adopt them and retain them.
- AI can be created to have virtually any goal, but almost any sufficiently ambitious goal can lead to subgoals of self-preservation, resource acquisition and curiosity to understand the world better—the former two may potentially lead a superintelligent AI to cause problems for humans, and the latter may prevent it from retaining the goals we give it.
- Although many broad ethical principles are agreed upon by most humans, it's unclear how to apply them to other entities, such as non-human animals and future AIs.
- It's unclear how to imbue a superintelligent AI with an ultimate goal that neither is undefined nor leads to the elimination of humanity, making it timely to rekindle research on some of the thorniest issues in philosophy!

Note | Page 284

David Chalmers, a famous Australian philosopher rarely seen without a playful smile and a black leather jacket. He followed his heart into philosophy despite making the finals at the International Mathematics Olympiad—and despite the fact that his only B grade in college, shattering his otherwise straight As, was for an introductory philosophy course.

Note | Page 289

Reminds me of sarah pondering a soft serve cone v baskin robbins cone as to which tasted like real ice cream, and what if...

Note | Page 294

"Although we're still nowhere near being able to measure every single firing of all of your roughly hundred billion neurons, brain-reading technology is advancing rapidly, involving techniques with intimidating names such as fMRI, EEG, MEG, ECoG, ePhys and fluorescent voltage sensing. fMRI, which stands for functional magnetic resonance imaging, measures the magnetic properties of hydrogen nuclei to make a 3-D map of your brain roughly every second, with millimeter resolution. EEG (electroencephalography) and MEG (magnetoencephalography) measure the electric and magnetic field outside your head to map your brain thousands of times per second, but with poor resolution, unable to distinguish features smaller than a few centimeters. If you're squeamish, you'll appreciate that these three techniques are all noninvasive. If you don't mind opening up your skull, you have additional options. ECoG (electrocorticography) involves placing say a hundred wires on the surface of your brain, while ePhys (electrophysiology) involves inserting microwires, which are sometimes thinner than a human hair, deep into the brain to record voltages from as many as a thousand simultaneous locations." So if you train ML on enough data it should be able to correlate the data with the corresponding behavior and reported conscious thoughts & feelings perhaps

Note | Page 297

your consciousness lives in the past, with Christof Koch estimating that it lags behind the outside world by about a quarter second. Intriguingly, you can often react to things faster than you can become conscious of them, which proves that the information processing in charge of your most rapid reactions must be unconscious

Note | Page 298

it takes longer for you to analyze images than sounds because it's more complicated—which is why Olympic races are started with a bang rather than with a visual cue.

Note | Page 300

just like solids, liquids and gases, I think consciousness is an emergent phenomenon, with properties above and beyond those of its particles.

Note | Page 300

if a substance is so viscous that it's rigid, we call it a solid, otherwise we call it a fluid. And if a fluid isn't compressible, we call it a liquid, otherwise we call it a gas or a plasma, depending on how well it conducts electricity.

Note | Page 300

The Italian neuroscientist Giulio Tononi has proposed one such quantity, which he calls the "integrated information," denoted by the Greek letter Φ (Phi), which basically measures how much different parts of a system know about each other (see figure 8.5).

Note | Page 302

integrated information theory (IIT). I'd been arguing for decades that consciousness is the way information feels when being processed in certain complex ways.¹⁸ IIT agrees with this and replaces my vague phrase "certain complex ways" by a precise definition: the information processing needs to be integrated, that is, Φ needs to be large. Giulio's argument for this is as powerful as it is simple: the conscious system needs to be integrated into a unified whole, because if it instead consisted of two independent parts, then they'd feel like two separate conscious entities rather than one. In other words, if a conscious part of a brain or computer can't communicate with the rest, then the rest can't be part of its subjective experience. Giulio and his collaborators have measured a simplified version of Φ by using EEG to measure the brain's response to magnetic stimulation. IIT is defined only for discrete systems that can be in a finite number of states

Note | Page 303

under what conditions will a blob of matter be able to do these four things? remember compute learn experience

Note | Page 303

how can consciousness feel so non-physical if it's in fact a physical phenomenon? How can it feel so independent of its physical substrate? I think it's because it is rather independent of its physical substrate, the stuff in which it is a pattern! We encountered many beautiful examples of substrate-independent patterns in chapter 2, including waves, memories and computations. We saw how they weren't merely more than their parts (emergent), but rather independent of their parts, taking on a life of their own. For example, we saw how a future simulated mind or computer-game character would have no way of knowing whether it ran on Windows, Mac OS, an Android phone or some other operating system, because it would be substrate-independent.

Note | Page 304

This leads to a radical idea that I really like: If consciousness is the way that information feels when it's processed in certain ways, then it must be substrate-independent; it's only the structure of the information processing that matters, not the structure of the matter doing the information processing. In other words, consciousness is substrate-independent twice over!

Note | Page 304

If the information processing itself obeys certain principles, it can give rise to the higher-level emergent phenomenon that we call consciousness. This places your conscious experience not one but two levels up from the matter. No wonder your mind feels non-physical!

Note | Page 304

consciousness, but here are four necessary conditions that I'd bet on and have explored in my research:

Principle Definition Information principle: A conscious system has substantial information-storage capacity. Dynamics principle: A conscious system has substantial information-processing capacity. Independence principle: A conscious system has substantial independence from the rest of the world. Integration principle: A conscious system cannot consist of nearly independent parts.

Note | Page 309

By measuring how long it takes neurons to transmit information across the brain, they could conclude that we experience no more than about ten conscious thoughts or perceptions per second,

Note | Page 309

First of all, the space of possible AI experiences is huge compared to what we humans can experience. We have one class of qualia for each of our senses, but AIs can have vastly more types of sensors and internal representations of information, so we must avoid the pitfall of assuming that being an AI necessarily feels similar to being a person. Second, a brain-sized artificial consciousness could have millions of times more experiences than us per second, since electromagnetic signals travel at the speed of light—millions of times faster than neuron signals. However, the larger the AI, the slower its global thoughts must be to allow information time to flow between all its parts, as we saw in chapter 4. We'd therefore expect an Earth-sized "Gaia" AI to have only about ten conscious experiences per second, like a human, and a galaxy-sized AI could have only one global thought every 100,000 years or so—

Note | Page 310

This really brings to a head the aforementioned controversy about whether parts of a conscious entity can be conscious too. IIT predicts not, which means that if a future astronomically large AI is conscious, then almost all its information processing is unconscious. This would mean that if a civilization of smaller AIs improves its communication abilities to the point that a single conscious hive mind emerges, their much faster individual consciousnesses are suddenly extinguished. If the IIT prediction is wrong, on the other hand, the hive mind can coexist with the panoply of smaller conscious minds. Indeed, one could even imagine a nested hierarchy of consciousnesses at all levels from microscopic to cosmic.

Note | Page 312

Decisions fall on a spectrum between two extremes: You know exactly why you made that particular choice. You have no idea why you made that particular choice—it felt like you chose randomly on a whim. Free-will discussions usually center around a struggle to reconcile our goal-oriented decision-making behavior with the laws of physics: if you're choosing between the following two explanations for what you did, then which one is correct: "I asked her on a date because I really liked her" or "My particles made me do it by moving according to the laws of physics"? But we saw in the last chapter that both are correct:

Note | Page 313

Their subjective experience of free will is simply how their computations feel from inside: they don't know the outcome of a computation until they've finished it. That's what it means to say that the computation is the decision.

Note | Page 313

let's not let those controversies distract us from the elephant in the room: there can be no positive experiences if there are no experiences at all, that is, if there's no consciousness. In other words, without consciousness, there can be no happiness, goodness, beauty, meaning or purpose—just an astronomical waste of space. This implies that when people ask about the meaning of life as if it were the job of our cosmos to give meaning to our existence, they're getting it backward: It's not our Universe giving meaning to conscious beings, but conscious beings giving meaning to our Universe. So the very first goal on our wish list for the future should be retaining (and hopefully expanding) biological and/or artificial consciousness in our cosmos, rather than driving it extinct.

Note | Page 314

we humans have often founded our self-worth on the idea of human exceptionalism: the conviction that we're the smartest entities on the planet and therefore unique and superior. The rise of AI will force us to abandon this and become more humble. But perhaps that's something we should do anyway: after all, clinging to hubristic notions of superiority over others (individuals, ethnic groups, species and so on) has caused awful problems in the past, and may be an idea ready for retirement. Indeed, human exceptionalism hasn't only caused grief in the past, but it also appears unnecessary for human flourishing: if we discover a peaceful extraterrestrial civilization far more advanced than us in science, art and everything else we care about, this presumably wouldn't prevent people from continuing to experience meaning and purpose in their lives. We could retain our families, friends and broader communities, and all activities that give us meaning and purpose, hopefully having lost nothing but arrogance.

Note | Page 314

Philosophers like to go Latin on this distinction, by contrasting sapience (the ability to think intelligently) with sentience (the ability to subjectively experience qualia). We humans have built our identity on being Homo sapiens, the smartest entities around. As we prepare to be humbled by ever smarter machines, I suggest that we rebrand ourselves as Homo sentiens!

Note | Page 315

THE BOTTOM LINE: • There's no undisputed definition of "consciousness." I use the broad and non-anthropocentric definition consciousness = subjective experience. • Whether AIs are conscious in that sense is what matters for the thorniest ethical and philosophical problems posed by the rise of AI: Can AIs suffer? Should they have rights? Is uploading a subjective suicide? Could a future cosmos teeming

with AIs be the ultimate zombie apocalypse? • The problem of understanding intelligence shouldn't be conflated with three separate problems of consciousness: the "pretty hard problem" of predicting which physical systems are conscious, the "even harder problem" of predicting qualia, and the "really hard problem" of why anything at all is conscious. • The "pretty hard problem" of consciousness is scientific, since a theory that predicts which of your brain processes are conscious is experimentally testable and falsifiable, while it's currently unclear how science could fully resolve the two harder problems. • Neuroscience experiments suggest that many behaviors and brain regions are unconscious, with much of our conscious experience representing an after-the-fact summary of vastly larger amounts of unconscious information. • Generalizing consciousness predictions from brains to machines requires a theory. Consciousness appears to require not a particular kind of particle or field, but a particular kind of information processing that's fairly autonomous and integrated, so that the whole system is rather autonomous but its parts aren't. • Consciousness might feel so non-physical because it's doubly substrate-independent: if consciousness is the way information feels when being processed in certain complex ways, then it's merely the structure of the information processing that matters, not the structure of the matter doing the information processing. • If artificial consciousness is possible, then the space of possible AI experiences is likely to be huge compared to what we humans can experience, spanning a vast spectrum of qualia and timescales—all sharing a feeling of having free will. • Since there can be no meaning without consciousness, it's not our Universe giving meaning to conscious beings, but conscious beings giving meaning to our Universe. • This suggests that as we humans prepare to be humbled by ever smarter machines, we take comfort mainly in being Homo sentiens, not Homo sapiens.

Note | Page 317

Future of Life Institute (FLI),

Note | Page 317

Foundational Questions Institute (see <http://fqxi.org>),

Note | Page 320

we started building machines that could outperform not only our muscles, but our minds as well.

Note | Page 322

the "Chatham House Rule," which prohibits participants from subsequently revealing who said what.*

Note | Page 327

This dramatic growth of AI-safety work wasn't limited to academia. Amazon, DeepMind, Facebook, Google, IBM and Microsoft launched an industry partnership for beneficial AI.8

Note | Page 329

The Asilomar AI Principles Artificial intelligence has already provided beneficial tools that are used every day by people around the world. Its continued development, guided by the following principles, will offer amazing opportunities to help and empower people in the decades and centuries ahead. If you too want to join as a signatory, you can do so here: <http://futureoflife.org/ai-principles>.

Note | Page 332

watch our long-term panel discussion on YouTube,¹¹ Elon Musk, Stuart Russell, Ray Kurzweil, Demis Hassabis, Sam Harris, Nick Bostrom, David Chalmers, Bart Selman and Jaan Tallinn all agreed that superintelligence would probably be developed and that safety research was important. Discussion panel on superintelligence with Elon Musk and other great minds: <http://tinyurl.com/asilomarAI>.

Note | Page 334

Erik Brynjolfsson spoke of two kinds of optimism in his Asilomar talk. First there's the unconditional kind, such as the positive expectation that the Sun will rise tomorrow morning. Then there's what he called "mindful optimism," which is the expectation that good things will happen if you plan carefully and work hard for them. That's the kind of optimism I now feel about the future of life.

Note | Page 334

Erik pointed out that according to game theory, positive visions form the foundation of a large fraction of all collaboration in the world, from marriages and corporate mergers to the decision of independent states to form the USA. After all, why sacrifice something you have if you can't imagine the even greater gain that this will provide? This means that we should be imagining positive futures not only for ourselves, but also for society and for humanity itself. In other words, we need more existential hope!

Note | Page 334

futuristic visions in literature and film are predominantly dystopian. In other words, we as a society are planning our future about as poorly as that hypothetical MIT student. This is why we need more mindful optimists. And

Note | Page 336

A strategy that's likely to help with essentially all AI challenges is for us to get our act together and improve our human society before AI fully takes off. We're better off educating our young to make technology robust and beneficial before ceding great power to it. We're better off modernizing our laws before technology makes them obsolete. We're better off resolving international conflicts before they escalate into an arms race in autonomous weapons. We're better off creating an economy that ensures prosperity for all before AI potentially amplifies inequalities. We're better off in a society where AI-safety research results get implemented rather than ignored. And looking further ahead, to challenges related to

superhuman AGI, we're better off agreeing on at least some basic ethical standards before we start teaching these standards to powerful machines.

Note | Page 336

In other words, one of the best ways for you to improve the future of life is to improve tomorrow.

Note | Page 336

Do you want to be someone who interrupts all their conversations by checking their smartphone, or someone who feels empowered by using technology in a planned and deliberate way? Do you want to own your technology or do you want your technology to own you?

Note | Page 336

We're the guardians of the future of life now as we shape the age of AI. Our future isn't written in stone and just waiting to happen to us—it's ours to create. Let's create an inspiring one together!

Note | Page 345

The study of neural correlates of consciousness (NCCs) has become quite mainstream in the neuroscience community in recent years—see, e.g., Geraint Rees, Gabriel Kreiman, and Christof Koch, “Neural Correlates of Consciousness in Humans,” *Nature Reviews Neuroscience* 3 (2002): 261–270, and Thomas Metzinger, *Neural Correlates of Consciousness: Empirical and Conceptual Questions* (Cambridge, MA: MIT Press, 2000).

Note | Page 347

For more about the Partnership on AI to benefit people and society, see their website: <https://www.partnershiponai.org>.

Note | Page 348

Among the principles that didn't make the final cut, one of my favorites was this one: “Consciousness caution: There being no consensus, we should avoid strong assumptions as to whether or not advanced AI would possess or require consciousness or feelings.” It went through many iterations, and in the last one, the controversial word “consciousness” was replaced by “subjective experience”—but this principle nonetheless got only 88% approval, just barely falling short of the 90% cutoff.
